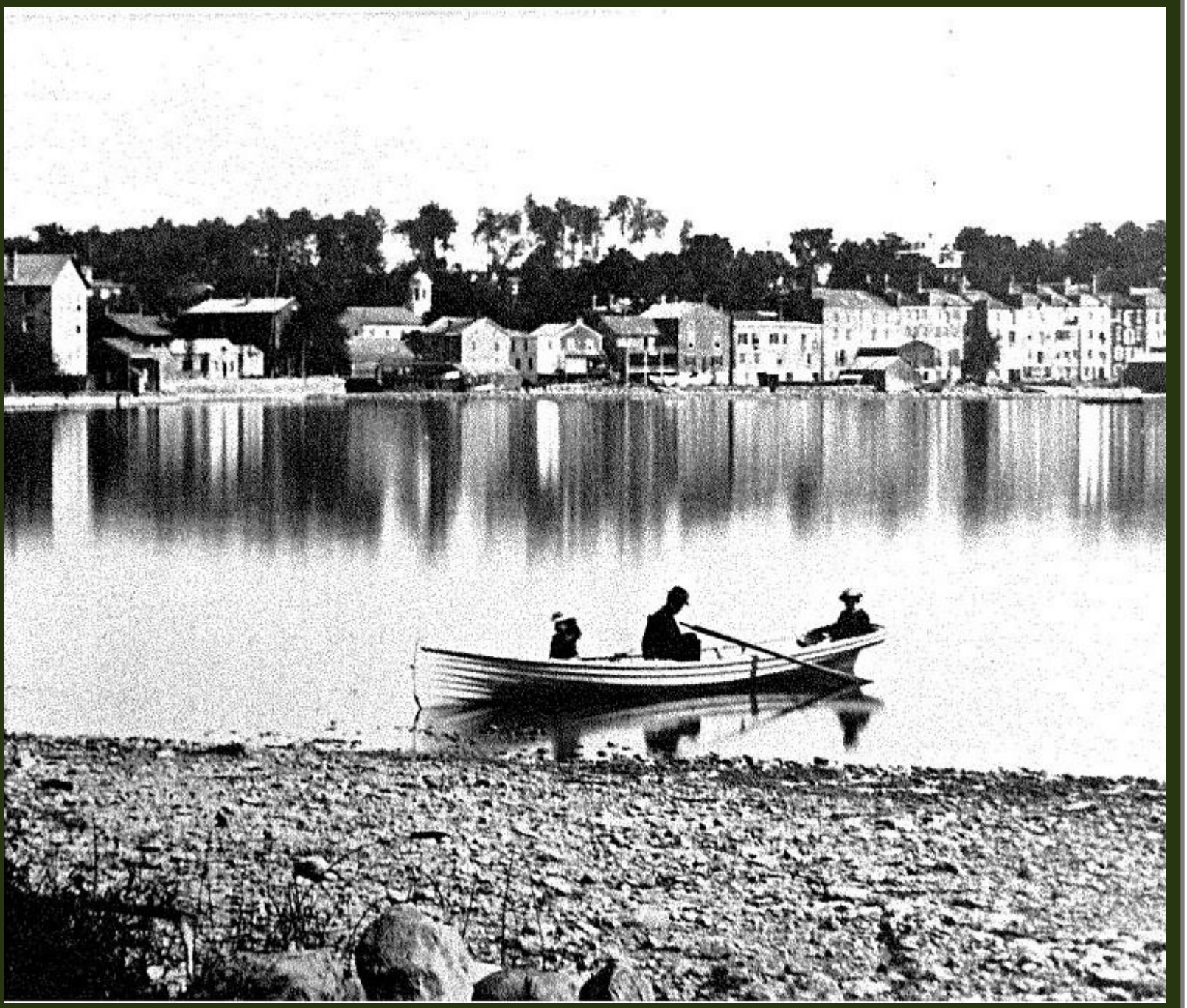
SKANEATELES LAKE AND WATERSHED 2025 ANNUAL REPORT

VOLUME LI



CITY OF SYRACUSE
DEPARTMENT OF WATER
WATER QUALITY MANAGEMENT

SKANEATELES LAKE AND WATERSHED
2025 ANNUAL REPORT

VOLUME LI

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Appendix

Appendix A – Exhibits

1. SKANEATELES LAKE AND WATERSHED

1.1 General Lake and Watershed Characteristics

Skaneateles Lake lies within the Oswego River Drainage Basin. It is the fourth largest and third deepest of the Finger Lakes (Exhibit A). It has a surface area of 13.6 square miles and has a maximum depth of 300 feet. It was formed during the Pleistocene Era by glacial scour and morainic damming. Thus, the lake basin is very steeply sloped with a small littoral zone and is "U" shaped. Approximately 80% of the lake's volume is below a depth of 30 feet.

The lake is dimictic, and complete turnover occurs twice per year (Spring and Fall). Strong stratification develops during the summer. In winter, a weaker inverse stratification develops where colder (and at times denser) water lies above warmer water. This occurs regardless of ice formation.

Skaneateles Lake is considered oligotrophic: it is low in nutrients and biological productivity. This is most likely a result of the small drainage basin to lake surface area ratio (4:1). Other features of oligotrophy include: (a) a deep, thick metalimnion resulting from greater light penetration, which forms during summer stratification, (b) an orthograde distribution of oxygen, also forming during summer stratification, (c) a blue-green color that demonstrates deep light penetration by the blue wavelengths due to the low concentrations of phytoplankton and humic substances, and (d) high transparency. The oligotrophic state of the lake makes it ideal for drinking water supply and recreation.

Much of the lake's shoreline is steeply sloped, especially at the southern end of the lake where cliffs can exceed 100 feet in height. Generally, the elevation is higher, and the slopes are steeper in the southern portion of the watershed as compared to the northern area (Exhibit A). The highest elevation is found in the southeastern portion of the watershed at 1,980 feet above sea level (USGS). Watershed acreage calculated by the Geographic Information System (GIS) is 37,724 acres or 58.94 square miles. The physical characteristics of the lake and drainage basin are summarized in Table No. 1.

Table 1 Skaneateles Lake and Drainage Basin Characteristics

Parameter	Value
Mean High Water Elevation*	863.27 Syracuse Datum
Mean High Water Elevation*	865.02 NVD
Length	15 Miles
Average Width	0.90 Miles
Maximum Depth	300 Feet
Mean Depth	145 Feet
Lake Surface Area	13.6 Square Miles
Drainage Basin Area (Land)	58.94 Square Miles
Drainage Basin: Lake Surface Ratio	4.33: 1.0
Lake Volume	412 Billion Gallons
Highest Elevation*	1,980 NVD

* Feet Above Sea Level

1.2 Watershed Soils

There are several soil associations within the watershed (Exhibit B and Table No. 2). The associations discussed below account for 80% of the watershed area and are considered representative of the character of the watershed. The remainder (20%) will not be discussed. It is noted that the associations generated by the GIS compare well, though not exactly, with the U.S. Department of Agriculture Soil Surveys prepared for each county. It would appear this is due to the inherent discrepancies often found between maps and mapping systems.

The major soil associations found in the northern two-thirds of the watershed are: Honeoye-Lima, Lansing-Conesus, Honeoye-Lansing, and Aurora-Angola-Darien. These associations are generally characterized as deep, medium textured silt loams and gravelly silt loams. Slopes range from 2-8%, except the Honeoye-Lansing Association that has slopes of 15-25%. They are moderately well, to well drained. Permeability is slow to moderate, and seasonal ground water is generally 15" to <36".

The Langford-Erie Association comprises the southwestern portion of the watershed (about 15% of the watershed). It is considered a deep, medium textured silt loam with slopes from 2% to 25%. The soils are poorly to well drained, and slowly to moderately permeable. A medium textured fragipan can be observed in some areas. Seasonal ground water and bedrock are shallow at 6" to 20", and 20" to 40", for the Langford and Erie series, respectively.

There are three (3) associations found primarily within Cortland County (southeastern portion of the watershed): Valois-Langford-Lansing, Volusia-Mardin-Lordstown, and Lordstown-Volusia-Mardin. This is equivalent to about 13% of the watershed. The soils are medium textured silt loams and gravelly silt loams. Slopes can be extreme and can exceed 55%. They are poorly to well drained and very slowly to moderately permeable. Seasonal ground water ranges from 18" to 60".

Typically, the soils of the watershed are gravelly to medium-textured silt loams that were formed in glacial till. Slopes are generally 2-25% but can exceed 55% in the southern portion of the watershed. The soils are poorly to moderately well drained, and very slowly to slowly permeable. The depth to seasonal ground water is generally <3.0 feet. All of the soils pose a severe risk of erosion if left bare, with increasing degree of slope compounding the potential for soil loss. The use of conventional onsite wastewater treatment systems (OWTS) is severely limited due to high seasonal ground water, degree of slope, and poor permeability.

Table 2 Soils of the Skaneateles Lake Watershed

Association	County	Percent of County	Dominant Location in Watershed	Percent of Watershed	Slope	Soil Type and Seasonal Ground Water	Drainage/ Permeability
Honeoye-Lima	Onondaga	39%	West & Northwest Onondaga	20%	2-8%	Deep, medium textured silt loams and gravelly silt loams Groundwater: 15-36"	Moderately well, to well drained and very slow to moderately permeable
Lansing-Conesus	Onondaga & Cayuga	17% and 22%	Spafford Uplands and Northeast Cayuga	16%	2-8%		
Honeoye-Lansing	Onondaga	9%	Spafford Shoreline	6%	15-25%		
Aurora-Angola-Darien	Onondaga	14%	North of Spafford Town Line	8%	2-8%		Poorly to moderately well drained
Langford-Erie	Cayuga	45%	South Cayuga	15%	2-25%	Deep, medium textured silt loams Groundwater: 6-20"	Poorly to well drained and slow to moderately permeable
Valois-Langford-Lansing	Cortland	87%	Lowlands	15%	2-55%	Medium textured and gravelly silt loams Groundwater: 18-60"	Poorly to well drained
Volusia-Mardin-Lordstown			Uplands				
Lordstown-Volusia-Mardin			Hewitt Forest				

1.3 Demographics, Land Use and Land Ownership

The watershed population totals approximately 4,158 people residing in 2,819 dwelling units (Table Nos. 3 and 4) [3]. Of the total number of dwelling units, approximately 1,065 are lakeshore dwellings. Land ownership is estimated as 52% private/residential (developed or vacant), 37% agricultural, 9% public and 2% commercial (Table No. 5). Based upon the GIS land use coverages, land use is calculated as: 48.2% agricultural; 40.3% open/forest; 5.4% residential; 4.7% brush; 0.9% other development; 0.3% ponds; 0.2% commercial (Table No. 6). Land use is defined as land cover acreage derived from the aerial orthophotographs.

Political subdivisions within the watershed include parts of three counties and seven townships. However, two of the towns, Marcellus and Owasco, account for less than 300 acres (or < 1.0% of the total watershed area). Onondaga County accounts for 51% of the watershed land area and 74% of the watershed population. Within the county are parts of the Towns of Skaneateles and Spafford, and the Village of Skaneateles. Forty-five percent (45%) of county lands are owned by the agricultural community. Of the estimated 2,148 dwelling units within the county, approximately 298 are connected to the Village of Skaneateles sanitary sewer system. The Village is the only municipality in the watershed having a public sewer system. The remaining watershed homes use Onsite Wastewater Treatment Systems (OWTS) or holding tanks for waste dispersal or collection (discussed in subsequent sections of this report). Several commercial buildings located on the south side of Route 20 in the village business district are within the watershed. All are connected to the municipal sanitary sewer system. Most buildings are 3-5 story structures. Various businesses occupy

the ground level floors of these buildings, and several apartments/inhabitants occupy the upper stories.

Cayuga County accounts for 34% of the land and 10% of the population within the watershed, which includes two towns, Niles and Sempronius. Agricultural lands account for 29% of the county land area. Most of the residential development in this county is on the lakeshore and is predominantly seasonal. Much of the shoreline is very steeply sloped.

Cortland County accounts for 15% of the watershed. The Town of Scott is the single town within this portion of the watershed. It has a population of 655 or approximately 16% of the total watershed population. Soil conditions and extreme topography of the area limit development. Farming accounts for 28% of the watershed land area.

Table 3 Skaneateles Watershed Population Distribution and Watershed Land Area by Town

Town	Dwelling Units	Population	Percent of Total Population	Percent of Watershed Land Area
Skaneateles	1,376	2,216	53.3%	26.0%
Spafford	689	843	20.3%	25.0%
Niles	365	371	8.9%	17.0%
Sempronius	76	73	1.8%	17.0%
Scott	313	655	15.8%	15.0%
Totals	2,819	4,158	100%	100.0%

Table 4 Skaneateles Watershed Population Distribution and Watershed Land Area by County

County	Dwelling Units	County Population	Percent of Watershed Population	Percent of Watershed Land Area	Density People/mile^2
Onondaga	2,148	3,059	73.6%	51%	74.3
Cayuga	441	444	10.7%	34%	20.3
Cortland	313	655	15.8%	15%	70.3
Totals	2,819	4,158	100%	100%	57.5*

* Average density

Table 5 Land Ownership in the Skaneateles Watershed

Land Ownership	Acres	Percent of Total
Agricultural	13,734	37%
Public	3,575	9%
Commercial	672	2%
Residential/Private	19,740	52%
Total	37,720	100%

Table 6 Land Use in the Skaneateles Watershed

Land Use	Acres	Percent of Total
Agricultural	16,398	43.5%
Commercial	58	0.2%
Residential	2,029	5.4%
Forest/Open	15,817	41.9%
Brush	1,792	4.7%
Pond/Wetland	1,556	4.1%
Other Development	322	0.2%
Total	37,972	100.0%

1.4 Skaneateles Lake Watershed Water Budget

The Skaneateles Lake Watershed receives an average 42.98" of precipitation per year. Of this amount, approximately 50% of the total precipitation produces runoff, or yield, to the lake [4,5]. Yield or runoff is the actual amount of water that reaches the lake because of precipitation. The yield coefficient (also called the hydrologic response coefficient) is an expression of the proportion of precipitation that reaches the lake.

Expressed as volume, annual precipitation produces approximately 53 billion gallons of water [6]. The runoff (26 billion gallons) provides the equivalent of about 9 feet of lake elevation (2.836 billion gallons/foot) [7]. Approximately one-third of the yield is precipitation that falls directly onto the lake surface, i.e., water that does not flow across or percolate through the soil before entering the lake. This is important to take into consideration when attempting to determine potential contaminant loading to the lake, since not all yield can be attributed to surface or sub-surface runoff. The remaining 26 billion gallons (50% of the total precipitation) is the net loss due to evapotranspiration, soil moisture recharge and groundwater recharge.

Of the total yield produced, the Water Department utilizes about 13 billion gallons per year for water supply to the city. The Village and Town of Skaneateles use approximately 0.26 billion gallons. About 10.5 billion gallons are discharged through the outlet of the lake to Skaneateles Creek. The latter is done to maintain elevations that satisfy the many uses of the lake: public and private water supply, storage for seasonal runoff, fishery spawning areas and recreation.

Three criteria are used to determine the rate of discharge through the lake's outlet for lake elevation management; (1): current levels as compared to the drawdown guideline levels, (2): current rates of precipitation, (3): the amount of water stored in the snowpack. For the latter, cores of snowpack at eighteen (18) locations within the watershed are analyzed weekly for water content.

1.5 Watershed Precipitation, Yield, Lake Level and Dam Discharges

The lake elevation on January 1 was 860.75, below the Monthly Low Drawdown Guideline Desired Range of 861.50. Snowfall for the month totaled 32.00" (av. 22.49"). The monthly elevation change was minimal, decreasing to 860.60 on February 1.

Recorded precipitation in February was above the monthly average at 3.08" (av. 2.63"). The lake level increased marginally (0.08') for the month to 860.65 on February 28. The lake level increased to 861.45 on March 31, just below the Monthly Low Drawdown Guideline Desired Range of 861.50. Rainfall in April totaled 4.17" (av. 3.61") resulting in a .75' monthly increase in lake elevation. Dam discharges for the months of January through April were maintained at a minimum.

Rainfall was recorded on twenty-five days in May, totaling 7.00" (av. 3.63") for the month, just below the highest recorded May monthly total of 7.24" recorded in 2000. Except for a 2.23" rain event on May 6, daily precipitation totals were all below 0.50". The lake elevation increased 0.90' in May from 862.20 on May 1 to 863.10 on June 1. Dam discharges averaged 92.53 MGD (av. 35.76 MGD) for the month, increasing from 6.92 MGD on May 6 to 138.23 MGD on May 31.

For the January-May period, precipitation totaled 17.51" (av. 15.87"). The yield and yield coefficient were 133.93 MGD (av. 118.68 MGD) and 0.75 (av. 0.89) respectively.

Snowfall for the 2024-25 season totaled 84.50" (av. 85.01"). A measurable snowpack was recorded on thirteen (weekly) sample dates from December 2024 through March 2025.

June's rainfall totaled of 4.58" (av. 4.02") Dam discharges averaged 132.84 MGD (av. 21.26 MGD), resulting in a monthly elevation decrease of 0.36'. July's precipitation total was 3.32" (av. 4.02"). The lake elevation decreased from 862.74 on July 1 to 862.20 on July 31. Dam discharges averaged 132.84 MGD (av. 20.69 MGD).

August precipitation total was below average at 3.25" (av. 4.00"). The monthly lake elevation decreased by 0.69' to 861.65 on August 31. Dam releases averaged 9.85 MGD (av. 13.59 MGD).

Precipitation for the June-August period totaled 11.15" (av. 12.03"). The yield and yield coefficient were above average at 44.19 MGD (av. 36.16 MGD) and 0.28 (av. 0.19) respectively.

The September 1 Lake elevation (861.62), was below the Monthly Low Drawdown Guideline Desired Range of 862.50, declining throughout the month to 860.95 on October 1. Monthly rainfall total was below average at 2.07" (av. 3.92"). The lake elevation continued to decrease through October to 860.43 on October 31. Total rainfall for October was 4.12" (av. 4.03"). Dam releases averaged 9.74 MGD (av. 13.13 MGD).

The lake level declined marginally in November (0.05'). Rainfall was below the monthly average at 2.85" (av. 3.65"). December's above average precipitation of 3.80" (av. 3.46") resulted in a monthly lake elevation increase of 0.12', from 860.28 on December 1 to 860.40 on December 31.

Precipitation for the September-December period totaled 12.83" (av. 15.06"). The yield for the period was 18.40 MGD (av. 47.70 MGD) and the yield coefficient was 0.14 (av. 0.28). Dam discharges for the period averaged 8.17 MGD (av. 20.11 MGD).

The Skaneateles Watershed received 41.50" of precipitation for the year (av. 42.98"). The average annual yield to the lake was 72.99 MGD (av. 74.39 MGD) with a yield coefficient of 0.44 (av. 0.49). Total discharge through the lake outlet for 2025 was 9,128 MG or 3.22' of elevation (av. 10,456 MG or 3.69', respectively). A summary of the above information can be found in Table No. 7.

Table 7 Skaneateles Watershed Data

Month	Precipitation in Inches		Yield in MGD		Elevation Feet Above Sea Level *;**		Dam Discharges in MGD	
	74 Year Average	2025	74 Year Average	2025	74 Year Average	2025	74 Year Average	2025
January	2.80	1.78	83.49	166.23	860.80	860.73	43.12	6.14
February	2.63	3.08	101.38	50.05	860.75	860.68	39.86	6.04
March	3.21	1.48	154.95	118.17	860.91	860.68	38.46	6.38
April	3.61	4.17	164.34	120.40	861.68	861.45	51.74	6.59
May	3.63	7.00	89.25	214.81	862.39	862.20	35.76	92.53
June	4.02	4.58	61.12	134.74	862.47	863.10	21.14	132.84
July	4.02	3.32	33.14	9.25	862.40	862.74	20.69	6.99
August	4.00	3.25	14.24	-11.43	862.02	862.31	13.59	9.85
September	3.92	2.07	11.82	-12.64	861.51	861.62	12.18	9.94
October	4.03	4.12	29.18	-7.68	861.02	860.95	13.13	9.74
November	3.65	2.85	60.03	42.20	860.71	860.33	21.50	7.12
December	3.46	3.80	89.78	51.72	860.66	860.28	43.77	5.91
Annual	**42.98**	**41.50**	**74.39**	**72.99**			**29.58**	**25.01**

* Syracuse Datum

** 1st Day of Month Lake Elevation

2. SAMPLING AND ANALYSIS

2.1 Turbidity Sampling and Analysis

The City of Syracuse Department of Water continued to operate under the guidelines of Title 10, Part 5 of the Official Compilation of Codes, Rules and Regulations of New York (10 NYCRR Part 5). All water quality criteria were met. Turbidity for each intake was recorded continuously using Hach 1720E turbidimeters (one per intake). The meters are calibrated routinely against readings taken on a Hach TU5200 model. The latter is calibrated every three months using the formazin method outlined in *Standards Methods for the Examination of Water and Wastewater, 23rd Edition.*

Readings were recorded by Water Plant personnel at four-hour intervals using the Hach TU5300 turbidimeter continuously recording turbidimeters. The results were included in the monthly reports submitted to the New York State Department of Health (NYSDOH) and are summarized in Table No. 8 of this report. The 2025 average turbidity readings for Intakes #1 and #2 were 0.49 NTU and 0.67 NTU, respectively.

Figure 1 illustrates relatively stable annual turbidity averages at Intake #1 and #2 through 2025. Following a significant spike in the September average from 2012 – 2014 (Figure 2), the trend line declined significantly through 2016 resulting in turbidity recordings consistent with the long-term average. September turbidity averages have fluctuated significantly from 2017 – 2025.

Daily maximum turbidity measurements at Intake #1 and #2 from 2004 through 2025 are shown in Figure 3. In 2025, the maximum turbidity for Intake #1 was 4.77 NTU.

Since monthly average data can mask spikes in turbidity, Figure 4 illustrates the number of occurrences with a daily maximum turbidity over 1 NTU for each month from 2004 – 2025. Daily

maximum turbidity above 1 NTU was recorded on 48 days in 2025 at Intake #1 (Figure 4 &Table No. 9).

Both Table No. 9 and Figure 4 display the variance in the number of days with maximum turbidity greater than 1 NTU at Intake #1 from 2004 to 2025.

Figure 1 2000-2025 Annual Turbidity Average

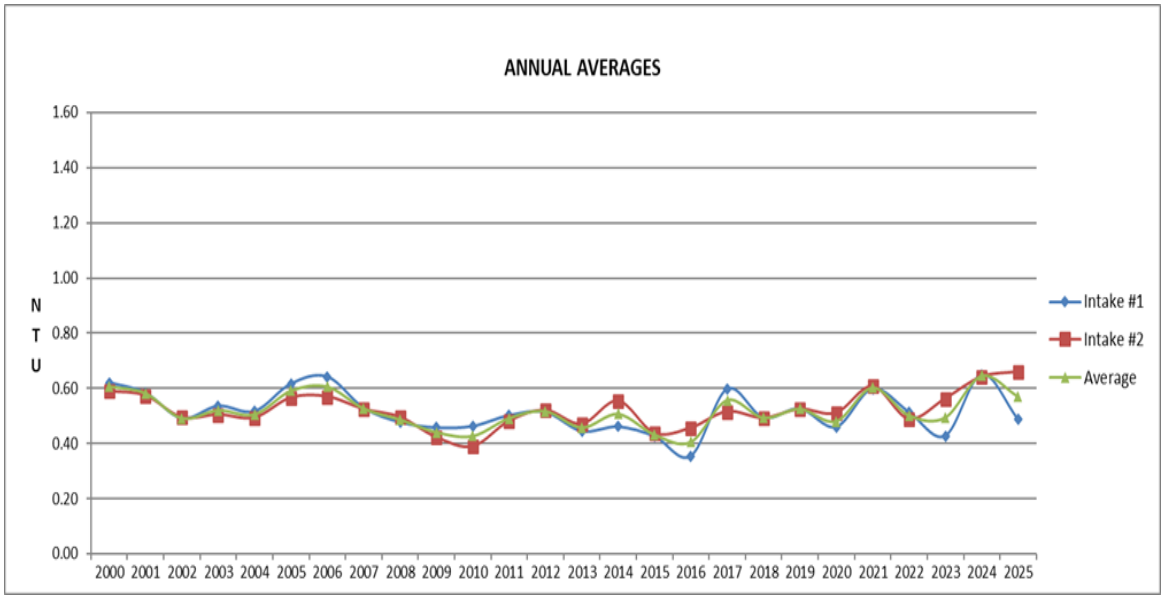


Figure 2 2000-2025 September Annual Turbidity Average

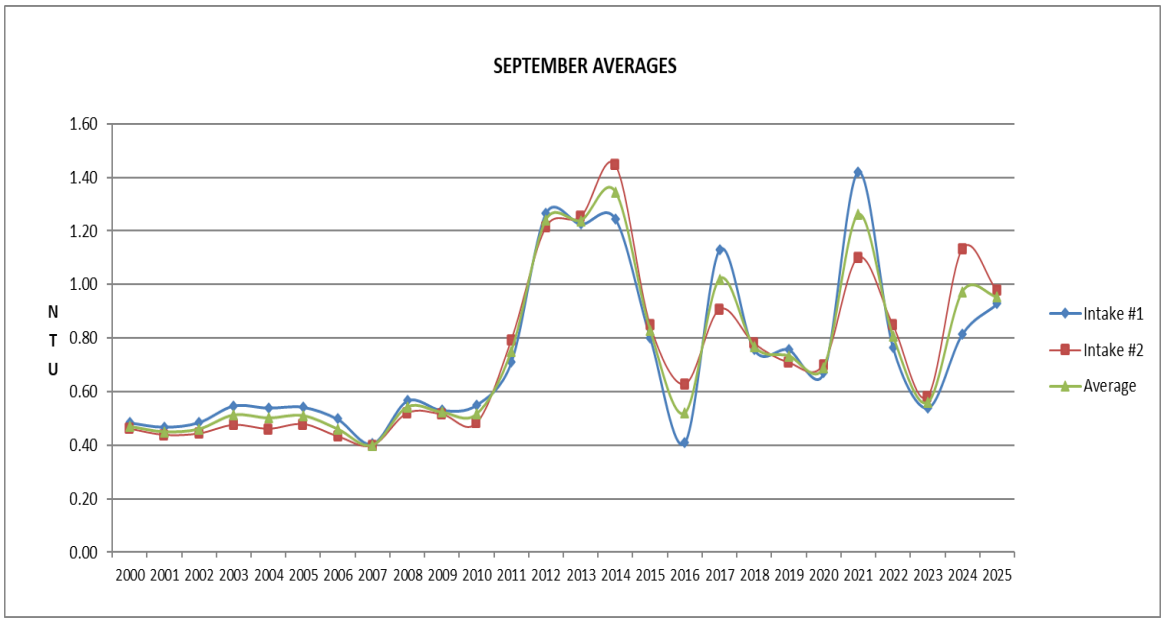


Figure 3 2004-2025 Daily Maximum Turbidity (NTU) Intake #1 & Intake #2

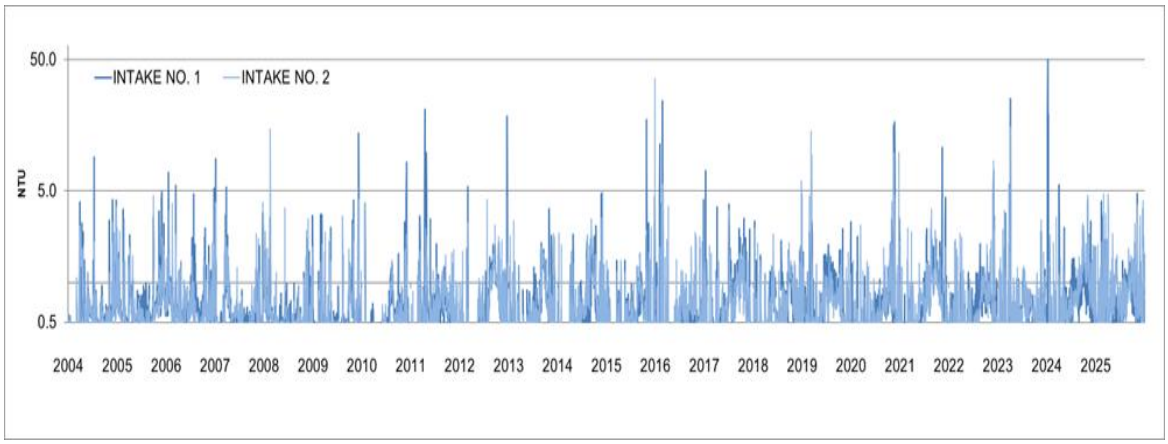


Figure 4 Monthly Counts of Turbidity >1 NTU at Intake 1, 2004-2025

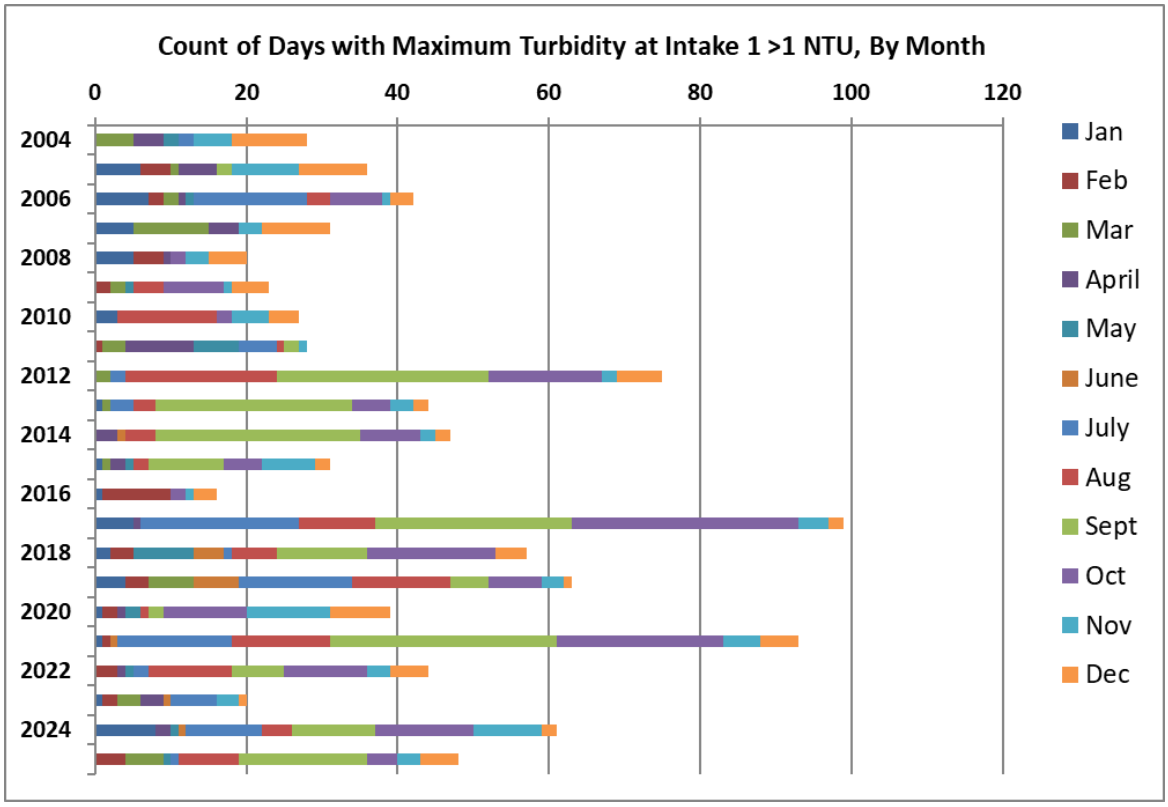


Table 8 Syracuse Water Plant Raw Water Maximum, Minimum, and Average Monthly Turbidity in NTU for 2025 (Both Intake #1 & #2)

Month	Intake 1 Maximum	Intake 1 Minimum	Intake 2 Maximum	Intake 2 Minimum	Intake 1 Average	Intake 2 Average
January	0.46	0.11	1.94	0.16	0.24	0.47
February	4.15	0.15	3.26	0.14	0.50	0.53
March	2.60	0.16	4.74	0.22	0.38	0.69
April	0.80	0.15	4.69	0.24	0.23	0.68
May	1.51	0.17	2.37	0.23	0.38	0.72
June	0.78	0.32	1.62	0.41	0.45	0.72
July	1.45	0.37	1.17	0.41	0.50	0.59
August	1.27	0.43	1.07	0.38	0.72	0.64
September	1.43	0.52	1.82	0.61	0.93	0.98
October	2.03	0.37	2.83	0.40	0.62	0.77
November	4.77	0.23	3.22	0.21	0.47	0.64
December	3.66	0.19	4.20	0.20	0.44	0.58
				Cumulative Average	**0.49**	**0.67**

Table 9 Number of Days with Maximum Turbidity > 1 NTU, Intake 1

	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	Summed by Month
Jan	0	6	7	5	5	0	3	0	0	1	0	1	1	5	2	4	1	1	0	1	8		51
Feb	0	4	2	0	4	2	0	1	0	0	0	0	9	0	3	3	2	1	3	2	0	4	40
Mar	5	1	2	10	0	2	0	3	2	1	0	1	0	0		6	0	0	0	3	0	5	41
April	4	5	1	4	1	0	0	9	0	0	3	2	0	1		0	1	0	1	3	2		37
May	2	0	1	0	0	1	0	6	0	0	0	1	0	0	8	0	2	0	1	0	1	1	24
June	0	0	0	0	0	0	0	0	0	0	1	0	0	0	4	6	0	1	0	1	1		14
July	2	0	15	0	0	0	0	5	2	3	0	0	0	21	1	15	0	15	2	6	10	1	98
Aug	0	0	3	0	0	4	13	1	20	3	4	2	0	10	6	13	1	13	11	0	4	8	116
Sept	0	2	0	0	0	0	0	2	28	26	27	10	0	26	12	5	2	30	7	0	11	17	205
Oct	0	0	7	0	2	8	2	0	15	5	8	5	2	30	17	7	11	22	11	0	13	4	169
Nov	5	9	1	3	3	1	5	1	2	3	2	7	1	4		3	11	5	3	3	9	3	84
Dec	10	9	3	9	5	5	4	0	6	2	2	2	3	2	4	1	8	5	5	1	2	5	93
Summed by Year	28	36	42	31	20	23	27	28	75	44	47	31	16	99	57	63	39	93	44	20	61	48	

2.2 Intake Closures

The Intakes may be closed to prevent a Treatment Technique Violation (TTV: any day with a turbidity reading exceeding 5.0 NTU) or Turbidity Event (a series of consecutive days during which at least one turbidity measurement each day exceeds 5.0 NTU).

Typically, Intake #2 is impacted by high wind events that re-suspend bottom sediments. This causes turbid water to enter the intake. Intake #1 may be impacted by significant runoff resulting from a high precipitation event. The high runoff causes sediment laden storm flows in Shotwell Brook. This brook is a main tributary on the north end of the lake that discharges approximately 1,200 feet south of Intake #1. When the wind speeds are high enough and the wind direction is southerly, this plume

of highly turbid water discharged into the lake can be transported over the intake. Most intake closures are to Intake #2.

Intake #2 was closed on 20 occasions in 2025 to prevent a TTV or a Turbidity Event. There were no Intake #1 closures related to turbidity concerns (Table No. 10).

Table 10 Raw Water Intake Closures

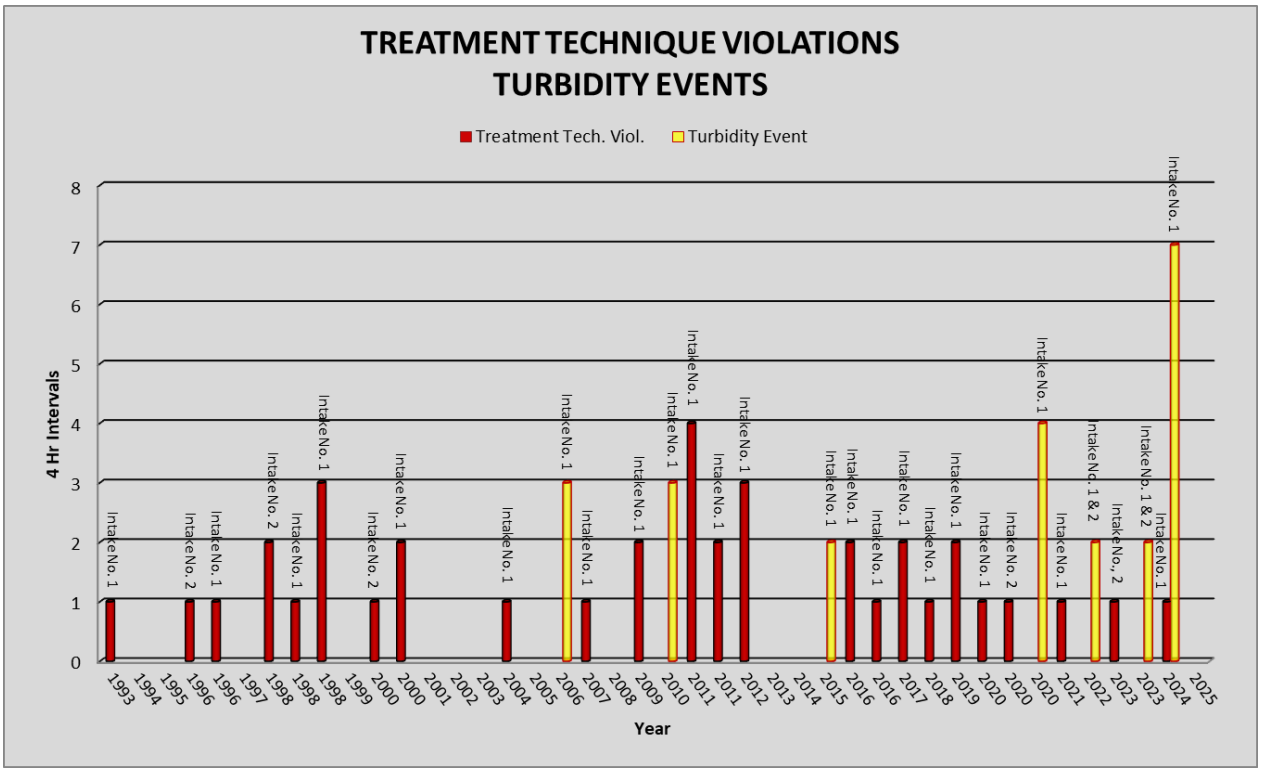
Date Intake Closed	Date Intake Opened	Intake Affected	Violation (Yes/No)
2/13/2025	2/15/2025	2	No
2/25/2025	2/26/2025	2	No
3/1/2025	3/2/2025	2	No
3/5/2025	3/6/2025	2	No
3/15/2025	3/17/2025	2	No
3/19/2025	3/20/2025	2	No
4/2/2025	4/3/2025	2	No
4/5/2025	4/6/2025	2	No
4/29/2025	4/30/2025	2	No
5/7/2025	5/8/2025	2	No
10/19/2025	10/21/2025	2	No
11/5/2025	11/6/2025	2	No
11/7/2025	11/9/2025	2	No
11/26/2025	11/27/2025	2	No
11/30/2025	12/1/2025	2	No
12/9/2025	12/11/2025	2	No
2/13/2025	2/15/2025	2	No
2/25/2025	2/26/2025	2	No

12/18/2025	12/21/2025	2	No
12/28/2025	12/30/2025	2	No

No TTVs or Turbidity Events occurred in 2025.

Refer to Figure 5 for TTV and Turbidity Events recorded since 1993. The Graph also displays the frequency that 5.0 NTU was exceeded (on a four-hour interval) during the TTV or Turbidity Event. Note that in 2015, 2022 and 2023, Turbidity Events consisted of only two 5.0 NTU exceedances at four-hour intervals, comparable to numerous TTV's illustrated in the graph. However, because the intervals were at 8:00 pm and 12:00 am, the conditions set forth by NYSDOH defining a Turbidity Event (a series of consecutive days during which at least one turbidity measurement each day exceeded 5.0 NTU) was attained.

Figure 5 1993-2025 Treatment Technique Violations vs Turbidity Events



2.3 Turbidity Analysis at City Reservoirs

Additional turbidity analyses were done Monday-Friday at the Woodland Reservoir and Westcott Reservoir effluent conduits, and four locations within the distribution system. These additional analyses were conducted by Department of Water, Maintenance and Operations Section personnel. Hach TU5200 turbidimeters are also located at the Woodland and Westcott Reservoir Gatehouses. The turbidimeters are also calibrated monthly by Water Plant personnel. During 2025 no monthly average turbidity in the distribution system exceeded 5.0 NTU. The results of this monitoring routine are included in the monthly reports sent to the NYSDOH.

2.4 Total and Fecal Coliform Sampling and Analyses

Total coliform and fecal coliform samples were collected at the Water Plant (raw water) five times/week/intake as directed by Title 10, Part 5 of the Official Compilation of Codes, Rules and Regulations of New York (10 NYCRR Part 5). The samples were analyzed by a NYSDOH certified commercial laboratory using the membrane filter (MF) technique. Of the 537 samples collected for total coliform analyses, there was one sample that exceeded 100 coliform forming units (cfu). Note that only the fecal coliform results are used to determine compliance with the City's filtration avoidance. Of the 537 samples collected for fecal coliform analyses, there were no samples that exceeded 20 cfu. No six-month threshold for fecal coliform density was exceeded. Table Nos. 11 and 12 summarize the monthly analyses for total and fecal coliform, for each intake, respectively.

In addition to the sampling conducted at the Water Treatment Plant, samples were routinely collected within the distribution system in Syracuse to ensure water quality. Samples were collected at 51 locations within the city. Approximately 197-239 samples were collected per month or 2,401 samples for the year.

Table 11 Skaneateles Lake Raw Water Total Coliform Analyses Rolling Six-month 10% Threshold for Part 5 Violation Criteria

For 6-Month Period Ending	Number of Samples Analyzed	Number of Samples Analyzed	Number of Samples >100 Colonies	Number of Samples >100 Colonies	Percent >100 Colonies	Percent >100 Colonies
Intake:	Number 1	Number 2	Number 1	Number 2	Number 1	Number 2
January	134	134	2	1	1.49%	0.75%
February	133	133	0	0	0.00%	0.00%
March	134	134	0	0	0.00%	0.00%
April	131	131	0	0	0.00%	0.00%
May	132	132	0	0	0.00%	0.00%
June	131	131	0	0	0.00%	0.00%
July	131	131	0	0	0.00%	0.00%
August	131	131	0	1	0.00%	0.76%
September	133	133	0	1	0.00%	0.75%
October	134	134	0	1	0.00%	0.75%
November	133	132	0	1	0.00%	0.76%
December	133	132	0	1	0.00%	0.76%

Table 12 Skaneateles Lake Raw Water Fecal Coliform Analyses Rolling Six-month 10% Threshold for Part 5 Violation Criteria

For 6-Month Period Ending	Number of Samples Analyzed	Number of Samples Analyzed	Number of Samples >20 Colonies	Number of Samples >20 Colonies	Percent >20 Colonies	Percent >20 Colonies
Intake:	Number 1	Number 2	Number 1	Number 2	Number 1	Number 2
January	134	134	0	1	0.00%	0.75%
February	133	133	0	1	0.00%	0.75%
March	134	134	0	1	0.00%	0.75%
April	131	131	0	1	0.00%	0.76%
May	132	132	0	1	0.00%	0.76%
June	131	131	0	0	0.00%	0.00%
July	131	131	0	0	0.00%	0.00%
August	131	131	0	0	0.00%	0.00%
September	133	133	0	0	0.00%	0.00%
October	134	134	0	0	0.00%	0.00%
November	133	132	0	0	0.00%	0.00%
December	133	132	0	0	0.00%	0.00%

2.5 Giardia and Cryptosporidium Sampling and Analyses

Analyses for the presence of Giardia cysts and Cryptosporidium oocysts began in 1985 and 1988, respectively. During 2025, samples were collected monthly from the raw water intakes (Intake #1 and Intake #2). A total of 24 samples were collected for Giardia and Cryptosporidium in 2025. Environmental Protection Agency (EPA) Method 1623 was the analysis utilized.

There were no confirmed Giardia cysts or Cryptosporidium oocysts detected in 2025. Since 1986, there have been 1,349 samples analyzed for Giardia and 1,337 samples for Cryptosporidium. Confirmed Giardia cysts have been observed in ten samples. Of these, eight were samples collected from tributaries within the watershed, one was a sample collected from Raw Water Intake #2, and one was collected from the Water Shop in 2003 (Table No. 13). Cryptosporidium oocysts have been observed on nine occasions since 1988. Of these, three sample locations were tributaries and three were from Raw Water Intake samples (Table No. 14).

Table 13 Skaneateles and Watershed *Giardia* Detection: Incidents of Confirmed Cysts

Incident Number	Date	Location	Cysts Detected	Calculated as Cysts/100 Liters
1	December 11, 1990	Grout Brook	2	2.6
2	March 14, 1991	Intake 2	1	0.1
3	March 14, 1991	Grout Brook	3	1.6
4	June 22, 1992	Shotwell Brook	2	6.6
5	June 22, 1992	One Mile Brook	4	4.2
6	August 28, 1992	Shotwell Brook	1	5.3
7	November 23, 1992	Willow Brook	1	0.3
8	November 23, 1992	Harrold Brook	2	2.1
9	March 24, 1993	Harrold Brook	1	1.1
10	May 13, 2003 *;**	Water Shop	1	2.0

* As of 2000, results are listed as cysts/liter

** As of August 2000, Method 1623 used for analyses

Table 14 Skaneateles and Watershed *Cryptosporidium* Detection: Incidents of Confirmed Oocysts

Incident Number	Date	Location	Oocysts Detected	Calculated as Oocysts/100 Liters
1	July 26, 1988	Intake 1	1	2.1
2	March 15. 1989	Intake 1	1	0.1
3	December 11, 1990	Grout Brook	2	2.6
4	November 23, 1992	Willow Brook	1	0.3
5	November 23, 1992	Harrold Brook	1	1.1
6	September 15, 2004 *;**	Intake 2 ***	2	4.0
7	September 21, 2011	Water Shop	5	0.1
8	October 3, 2011	Water Shop	1	0.02
9	October 3, 2011	Woodland Reservoir	2	0.01

* As of 2000, results are listed as cysts/liter ** As of August 2000, Method 1623 is used for analyses

*** Previous reports list Intake 1 as the affected intake. That was incorrect. The oocysts were in a sample from Intake 2

2.6 Skaneateles Lake Limnological Sampling and Analyses

2.6.1 Algal Analyses

The sampling regime included station samples and depth profile samples. The station sampling consists of collecting one-liter samples at eight locations on the lake at a depth of 20'. The sites are approximately two miles apart. Sampling was conducted on ten dates with 80 one-liter samples being collected and analyzed for algal content.

It is noted that the Secchi disk readings recorded during the station sampling can be somewhat skewed. Since stations I and VIII are shallow, the readings are consistently at the bottom, which is usually about six meters (as a result, samples are collected at about 15 feet). It is not unusual for the readings at the other Stations to vary considerably throughout the spring, summer and fall seasons, ranging from four to 16 meters (Figure 6). Figures 7 and 8 illustrate Secchi disk depth variations

between early summer and fall. References to Secchi disk readings in the narrative below are adjusted, i.e., only Stations II – VII were used to determine averages.

Depth profile samples were collected at a location approximately four miles from the northern shore in the center of the lake. On each of the six sampling dates, 21 one-liter samples were collected at 10-foot intervals from the surface to a depth of 200 feet. Each was analyzed for temperature and turbidity. Algal content was determined on those samples representing the different layers of stratification: six of the 21 samples collected on each date were analyzed for algal content. Thirty-six one-liter samples were analyzed for algal concentration during the season.

Sampling was conducted from May through October 2025. Most genera observed were from the Phyla Cyanophyta and Chrysophyta (Table No. 15). For Station Samples, four genera accounted for 83.27% of the total cell count: Polycystis (44.82%), Cyclotella (16.33%), Synedra (12.92%) and Cyanobium (9.20%). For Depth Profile samples, five genera accounted for 90.41% of the total cell count. Polycystis and Cyclotella were the dominant forms at 46.69% and 21.96% respectively. Synedra, Chroococcus Type I and Cyanobium accounted for 7.48%, 7.28% and 7.00% respectively.

The station and depth profile sample collection and analyses are discussed in greater detail below. Table No. 16 lists the dates and results of the Station analyses. Table No. 17 is a summary of the Depth Profile analyses.

Note: There have been changes to the identification (or naming) of the various genera over the past several years. For the most part, the changes apply to growth in the city reservoirs. A complete summary can be found in the draft document: City of Syracuse Water Department: Woodland Reservoir Treatment Procedures for Algal Control.

Figure 6 Average Secchi Disk Depths 1987-2025

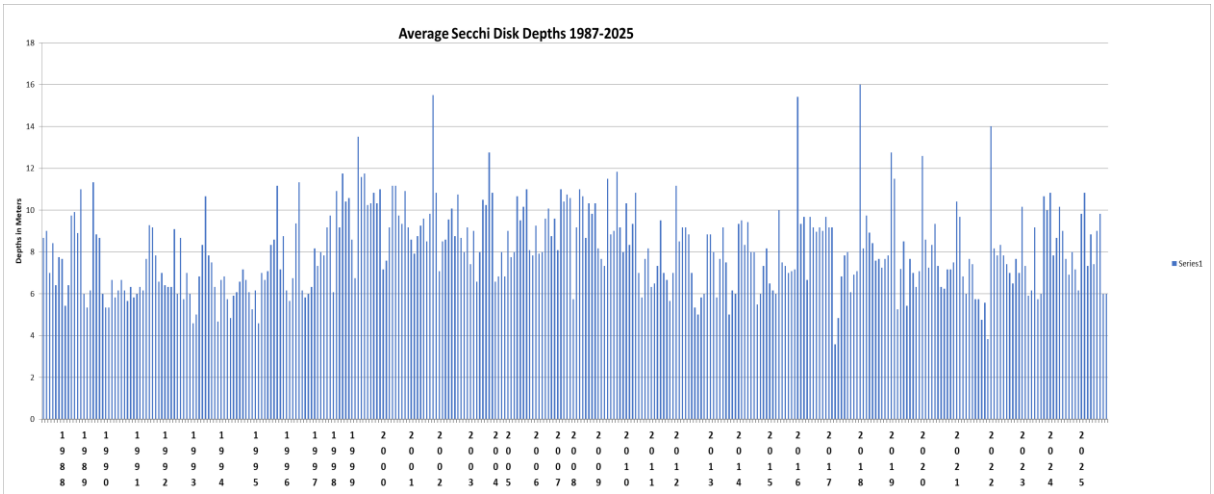


Figure 7 Average Secchi Disk Depths (June) 1987-2025

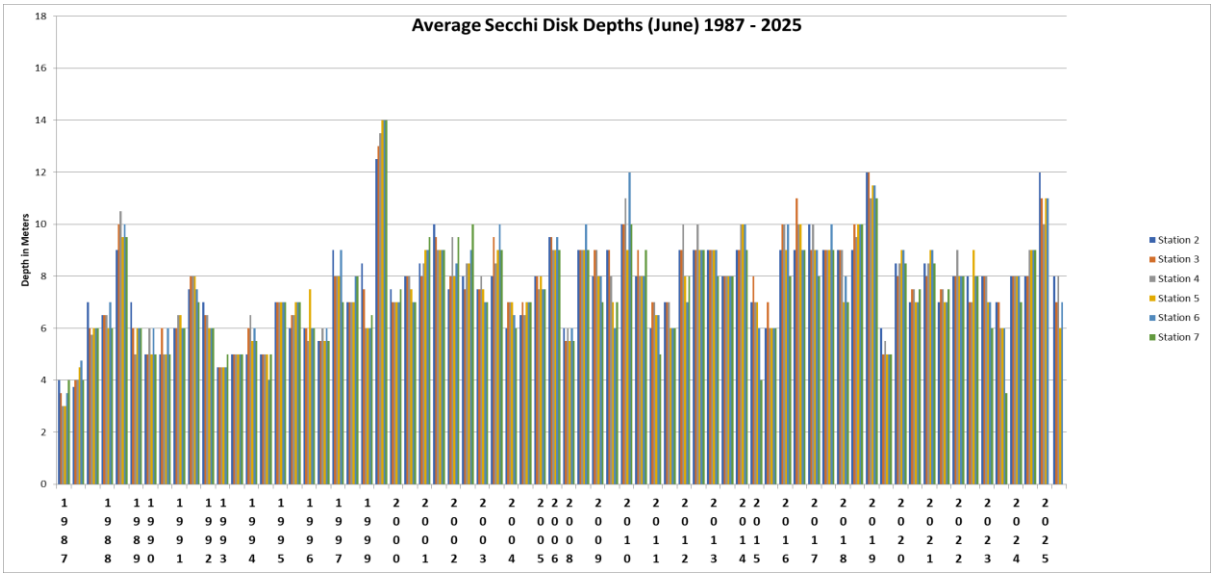


Figure 8 Average Secchi Disk Depths (September) 1987-2025

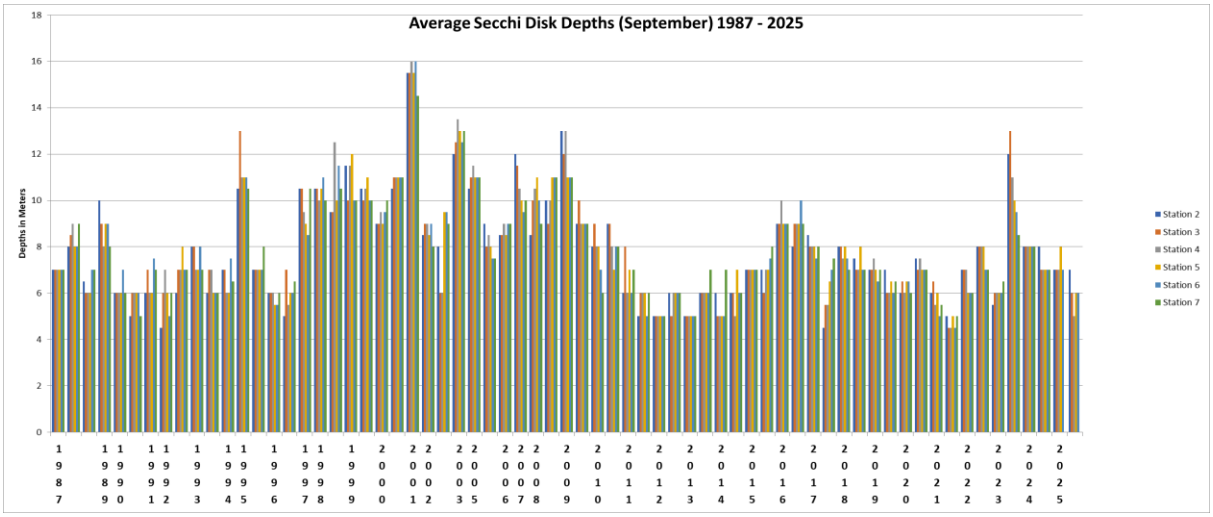


Table 15 Genera of Phytoplankton Typically Found in Skaneateles Lake

Phylum: Chrysophyta (Diatoms)

Achnanthes	Coscinodiscus	Diatoma	Meridion	Pinnularia	Surirella
Asterionella	Cyclotella	Fragellaria	Navicula	Stauroneis	Synedra
Cocconeis	Cymbella	Gomphonema	Nitzschia	Stephanodiscus	Tabellaria

Phylum: Chrysophyta (Golden-Brown)

Centritractus	Tribonema
Dinobryon	Mallomonas

Phylum: Chlorophyta (Green)

Botyrococcus	Cosmarium	Palmella	Straurastrum
Chlorcoccum	Hydrodictyon	Pediastrum	Tetraedron
Chlorella	Nitella	Phytoconis	Ulothrix
Coelastrum	Oocystis	Scenedesmus	Volvox

Phylum: Cyanophyta (Blue-Green)

Anabaena	Cyanarcus	Merismopedia
Aphanothece	Gomphosphaeria	Rivularia
Polycystis	Lyngbya	Oscillatoria
Chroococcus	Cyanobium	

Phylum: Euglenophyta (Flagellates)

Euglena	Trachelomonas

Phylum: Pyrrophyta (Dinoflagellates)

Ceratium	Peridinium

Table 16 Station Sampling Analyses Skaneateles Lake Algal Content 2025

Date	Average Total Cell Count	Dominant Form	Percent of Total	Average Secchi Disk Meters	Average Water Temperature ^{0}F
May 27	286	Polycystis 83.5%	83.5%	9.8	49.9
June 5	524	Polycystis 77.7% Synedra 11.5%	89.2%	10.8	53.8
June 25	1120	Polycystis 33.2% Cyclotella 29.5% Achnanthes 16.1% Cyclotella 12.5%	91.3%	7.4	65.9
July 10	1245	Polycystis 42.7% Cyclotella 26.9% Cyanobium 11.0% Synedra 9.2% Dinobryon 7.4%	97.2%	8.8	72.5
July 23	1413	Polycystis 35.1% Dinobryon 26.2% Synedra 21.9% Cyclotella 9.3%	92.5%	7.4	73.8
August 6	1190	Synedra 31.4% Cyclotella 30.3% Polycystis 23.5% Dinobryon 7.7%	92.9%	9.0	75.3
August 22	994	Polycystis 58.3% Cyclotella 20.5% Achnanthes 9.2%	88.0%	9.4	73.1
September 11	483	Polycystis 68.3% Cyclotella 9.4% Chroococcus Type I 7.7%	85.4%	7.2	68.3
September 26	246	Polycystis 68.3% Cyclotella 9.4% Chroococcus Type I 7.7%	85.4%	6.0	67.5
October 17	301	Achnanthes 42.5% Polycystis 34.6% Cyclotella 10.5%	87.6%	6.0	67.5
Average				**8.2**	**66.8**

Table 17 Depth Profile Sampling Analyses Skaneateles Lake Algal Content 2025

Date	Average Total Cell Count	Dominant Form	Percent of Total	Average Turbidity	Average Water Temperature ^{0}F
May 16	419	Polycystis 72.5% Achnanthes 17.8% Cyanobium 7.5%	97.8%	0.54	45.60
June 18	400	Polycystis 62.1% Synedra 23.4% Chroococcus Type I 4.3% Cyanobium 4.3%	94.1%	0.51	51.20
July 17	1684	Cyclotella 41.4% Polycystis 31.0% Dinobryon 7.0% Synedra 6.6%	86.0%	0.40	59.10
August 14	939	Polycystis 51.1% Cyanobium 14.7% Cyclotella 18.0-% Synedra 10.4%	94.2%	0.41	60.60
September 18	809	Polycystis 45.3% Chroococcus Type I 28.9% Cyclotella 14.7%	88.9%	0.54	56.30
October 28	272	Polycystis 70.1% Achnanthes 13.3% Cyanobium 5.6%	89.0%	0.60	51.30
Average	**754**			**0.50**	**54.02**

2.6.2 Station Sample Analysis

Total cell counts for station samples collected on May 27 averaged 286 cells/mL. Polycystis was the dominant form accounting for 83.5% of the total. Fourteen additional forms were identified, however each accounted for less than 4% of the total. Secchi disc readings averaged 9.8 meters. Water temperatures averaged 49.90 degrees Fahrenheit.

The total monthly cell counts for June averaged 822 cells/mL. On June 5 the total average cell count was 524 cells/mL. Polycystis and Synedra were dominant, accounting for 77.7% and 11.5% of the total, respectively. The June 25 total cell count increased significantly, averaging 1,120 cells/mL. Polycystis and Cyanobium were dominant, accounting for 33.2% and 29.5% of the total respectively. Achnanthes accounted for 16.1% and Cyclotella accounted for 12.5%. Secchi disk readings averaged 10.8 meters on June 5, declining significantly to 7.4 meters on June 25. The average water temperature recordings increased between sample events from 53.80 degrees Fahrenheit (June 5) to 65.90 degrees Fahrenheit (June 25).

The July monthly average total cell count was 1,329 cells/mL. On July 10, the total average cell count was 1,245 cells/mL. Polycystis was the dominant form, accounting for 42.7% of the total cell count. Cyclotella, Cyanobium and Synedra accounted for 26.9%, 11.0% and 9.2%, respectively. The July 23 average total cell count increased to 1,413 cells/mL. Polycystis, Dinobryon, Synedra and Cyclotella were dominant forms accounting for 35.1%, 26.2%, 21.9% and 9.3% of the total cell count, respectively. The average water temperature for the two sample events was 73.10 degrees Fahrenheit. Secchi disk readings averaged 8.1 meters.

The total average cell counts on August 6 decreased from July sample events, averaging 1,190 cells/mL. The dominant forms were Synedra (31.4%), Cyclotella (30.3%), Polycystis (23.5%) and Dinobryon (7.7%). On August 22, the average total cell count decreased to 994 cells/mL. Polycystis, Cyclotella and Achnanthes were dominant, accounting for 58.3%, 20.5% and 9.2%, respectively. For the month of August, water temperatures averaged 74.20 degrees Fahrenheit and Secchi disk readings averaged 9.2 meters.

Total cell counts for September 11 and September 26 averaged 483 cells/mL. and 246 cells/mL., respectively. For the September 11 sample event, dominant forms included Polycystis (68.3%), Cyclotella (9.4%), and Chroococcus Type I (7.7%). Polycystis was the dominant form for the September 26 event totaling 64.4%. Cyanobium and Achnanthes totaled 17.8% and 7.0%, respectively. Water temperatures for the month of September averaged 67.90 degrees Fahrenheit. Secchi disk readings averaged 6.6 meters.

Total cell counts for October 17 averaged 301 cells/mL. Achnanthes and Polycystis were the dominant forms accounting for 42.3% and 34.6% of the total cell count, respectively. Cyclotella accounted for 10.5%. Water temperatures averaged 67.50 degrees Fahrenheit and Secchi disc readings averaged 6.0 meters.

2.6.3 Depth Profile Sample Analysis

The total cell counts for the May 16 depth profile averaged 419 cells/mL. Water temperatures averaged 45.60 degrees Fahrenheit, ranging from 50.00 degrees Fahrenheit at the surface, declining to 43.00 degrees Fahrenheit (160’). Polycystis and Achnanthes were dominant at 72.5% and 17.8% of the total cell count. Cyanobium accounted for 7.5% of the total. The lake was exhibiting spring turnover, indicated by weak thermal stratification.

The total cell counts for the June 18 depth profile averaged 400 cells/mL. Polycystis was the dominant form, accounting for 62.1% of the total cell count. Synedra accounted for 23.4% and Chroococcus Type I and Cyanobium each accounted for 4.3% of the total cell count. The surface temperature increased to 62.00 degrees Fahrenheit; spring turnover was still evident with no defined epilimnion.

The July 17 depth profile total cell count average increased significantly from the June average to 1684 cells/mL. The dominant forms were Cyclotella (41.4%) and Polycystis (31.0%). Dinobryon and Synedra accounted for 7.0% and 6.6%, respectively. The lake surface temperature increased to 79.00 degrees Fahrenheit. The epilimnion was still forming, the metalimnion was shallow extending to 70’ and the hypolimnion was narrowly established from 70’ downward.

On August 14 the depth profile cell counts averaged 939 cells/mL. Polycystis was the dominant form accounting for 51.1% of the total count. Additional dominant forms included: Cyclotella (18.0%), Cyanobium (14.7%) and Synedra (10.4%). The surface water temperature remained warm at 78.00 degrees Fahrenheit. Thermal stratification was evident; the epilimnion was shallow, extending to 20’, the thermocline extended to 90’ and the hypolimnion from 90’ downward.

The September 18 depth profile average total cell count was 809 cells/mL. Polycystis was the dominant form comprising 45.3% of the total cell count. Chroococcus Type I and Cyclotella

accounted for 28.9% and 14.7% of the sample, respectively. The epilimnion extended to a depth of 40', the metalimnion extended to 80' and the hypolimnion from 80' downward. The lake's surface temperature cooled significantly to 71.00 degrees Fahrenheit.

The average total cell count on the October 28 sample event was 272 cells/mL. Polycystis was the dominant form at 70.1%. Achnanthes and Cyanobium accounted for 13.3% and 5.6% of the sample, respectively. The epilimnion extended to 80', the metalimnion to 90 ' and hypolimnion was extended from 90' downward.

2.7 Skaneateles Lake Algal Blooms

Algal blooms observed by the Skaneateles Lake Watershed Protection staff in 2025 were localized and limited to near-shore areas. Microcystin was detected on 17 dates in raw water samples collected from the water intakes in 2025. Microcystin was reported at or above 0.30 ug/L on 12 sample events for Intake #1 and 14 sample events for Intake #2 (Table No. 18).

Monitoring, identifying, sampling, and reporting algae blooms involved a collaborative effort between the Skaneateles Lake Association (SLA) Shoreline HABs Program and the City of Syracuse Water Department. The SLA Shoreline HABs Program comprising of select volunteers continued to monitor 25 zones around the perimeter of Skaneateles Lake in 2025. Suspicious algal blooms were reported to the NYSDEC Division of Water website HABsInfo@dec.ny.gov. Syracuse Water Department personnel routinely collected surface water skim samples following reports of suspicious blooms. Algal forms were identified, and cell counts performed on one-liter centrifuged samples under light microscopy to determine dominant algal forms.

As a result of the numerous volunteers and professional staff monitoring Skaneateles Lake and the heightened awareness of lakefront property owners and watercraft operators, the lake was intensely monitored, and lake conditions assessed in a timely manner. The frequent monitoring and the lake-wide surveillance program were instrumental in the early detection of algal blooms.

2.8 HAB Action Plan for Managing Microcystin in Skaneateles Lake

In coordination with the NYSDOH, a HAB Action Plan was developed in 2018 to ensure that the City's drinking water remained of high quality and microcystin was not detected in treated water above 0.3 µg/L. The Action Plan includes short-term and long-term measures and is updated annually.

2.8.1 Short Term Measures

The short-term measures provided for an aggressive monitoring program. Microcystin sampling at the City's drinking water intakes was initiated on July 7 and extended through October 22 of 2025. Raw water is analyzed weekly unless significant blooms are identified within the North basin or microcystin is detected in either raw or treated samples, triggering a more frequent sampling schedule. Samples were collected and transported to an ELAP certified lab on 33 occasions for analysis in 2025. Twenty-six raw water samples were reported with microcystin levels at or above the limit of quantitation (LOQ) of 0.3 µg/L (Table No. 18).

Table 18 Skaneateles Lake Microcystin Levels (ug/L) July 7 – October 22, 2025

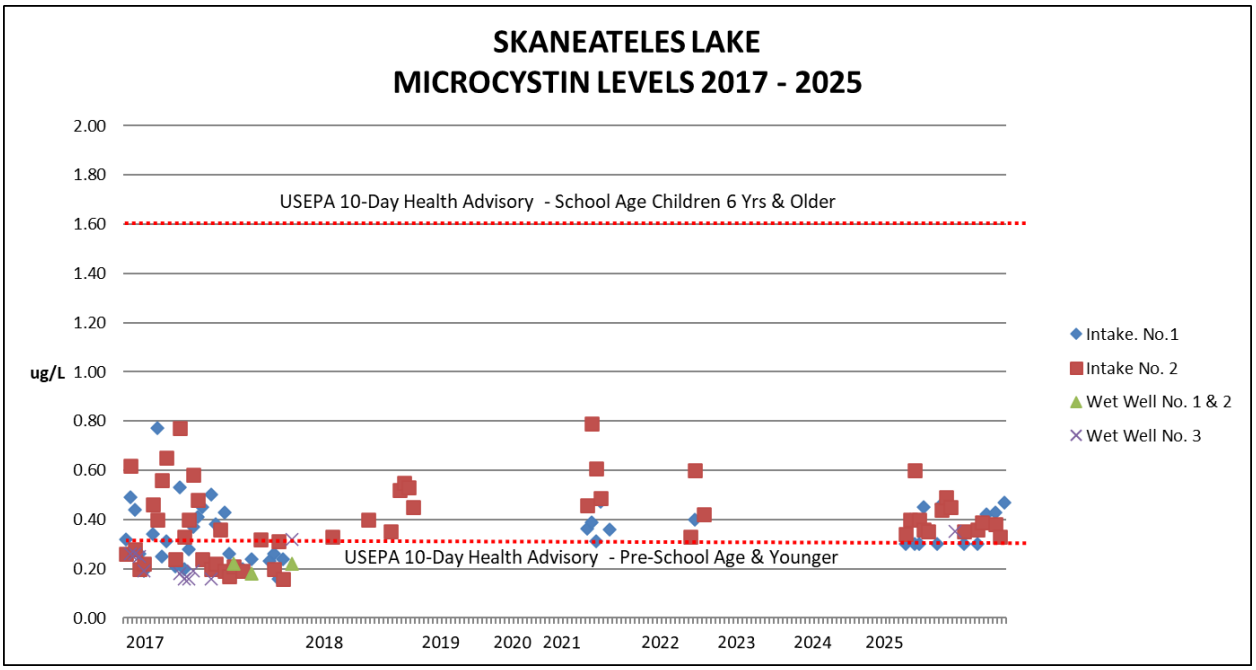
Date Sampled	7/7	7/14	7/21	7/28	8/4	8/11	8/18	8/25	9/2	9/8	9/11
Intake 1	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.30	0.40
Intake 2	ND	ND	ND	ND	ND	ND	ND	ND	ND	0.34	0.40
Clear Well 1 & 2	-	-	-	-	-	-	-	-	-	-	-
Clear Well 3	-	-	-	-	-	-	-	-	-	-	-
Skaneateles UV plant	-	-	-	-	-	-	-	-	-	-	-
Elbridge UV plant	-	-	-	-	-	-	-	-	-	-	-
Westcott incoming	-	-	-	-	-	-	-	-	-	-	-
Westcott outgoing	-	-	-	-	-	-	-	-	-	-	-
Woodland incoming	-	-	-	-	-	-	-	-	-	-	-
Wooland outgoing	-	-	-	-	-	-	-	-	-	-	-
Skaneateles HS	-	-	-	-	-	-	-	-	-	-	-
Byrne Dairy	-	-	-	-	-	-	-	-	-	-	-
Elbridge North	-	-	-	-	-	-	-	-	-	-	-
Jordan Town Hall	-	-	-	-	-	-	-	-	-	-	-
Syracuse Burnet Ave	-	-	-	-	-	-	-	-	-	-	-

Date Sampled	9/12	9/13	9/14	9/15	9/16	9/17	9/18	9/19	9/20	9/21	9/22
Intake 1	0.30	0.30	0.45	ND	ND	0.30	0.46	ND	ND	ND	ND
Intake 2	0.60	0.40	0.36	0.35	ND	ND	0.44	0.49	0.45	ND	ND
Clear Well 1 & 2	-	-	-	-	-	-	-	-	ND	ND	ND
Clear Well 3	-	-	-	-	-	-	-	-	0.35	ND	ND
Intake No. 2 Extension	ND	-	-	ND	-	ND	-	-	-	-	-
Skaneateles UV plant	-	-	-	-	-	-	-	-	-	-	-
Elbridge UV plant	-	-	-	-	-	-	-	-	-	-	-
Westcott incoming	-	-	-	-	-	-	-	-	-	-	-
Westcott outgoing	-	-	-	-	-	-	-	-	-	-	-
Woodland incoming	-	-	-	-	-	-	-	-	-	-	-
Wooland outgoing	-	-	-	-	-	-	-	-	-	-	-
Skaneateles HS	-	-	-	-	-	-	-	-	-	-	-
Byrne Dairy	-	-	-	-	-	-	-	-	-	-	-
Elbridge North	-	-	-	-	-	-	-	-	-	-	-
Jordan Town Hall	-	-	-	-	-	-	-	-	-	-	-
Syracuse Burnet Ave	-	-	-	-	-	-	-	-	-	-	-

Date Sampled	9/23	9/25	9/26	9/29	10/1	10/6	10/8	10/13	10/15	10/20	10/22
Intake 1	0.30	0.30	ND	0.42	ND	0.43	ND	0.47	ND	ND	ND
Intake 2	0.35	0.36	0.39	ND	ND	0.38	0.33	ND	ND	ND	ND
Clear Well 1 & 2	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
Clear Well 3	ND	ND	ND	ND	ND	ND	ND	ND	ND		ND
Skaneateles UV plant	-	-	-	-	-	-	-	-	-	-	-
Elbridge UV plant	-	-	-	-	-	-	-	-	-	-	-
Westcott incoming	-	-	-	-	-	-	-	-	-	-	-
Westcott outgoing	-	-	-	-	-	-	-	-	-	-	-
Woodland incoming	-	-	-	-	-	-	-	-	-	-	-
Wooland outgoing	-	-	-	-	-	-	-	-	-	-	-
Skaneateles HS	-	-	-	-	-	-	-	-	-	-	-
Byrne Dairy	-	-	-	-	-	-	-	-	-	-	-
Elbridge North	-	-	-	-	-	-	-	-	-	-	-
Jordan Town Hall	-	-	-	-	-	-	-	-	-	-	-
Syracuse Burnet Ave	-	-	-	-	-	-	-	-	-	-	-

Throughout periods of microcystin detections in raw water intakes the city has demonstrated that by boosting chlorine levels at the water intake cribs and wet wells and maintaining an elevated chlorine concentration in finished water, microcystin detection within the water distribution system could be avoided. Figure No. 9 shows the number of microcystin detections for both raw and treated water samples and the range of detections for the past nine years. Note that microcystin was not detected in 2019, 2020, 2023 and 2024.

Figure 9 Skaneateles Lake Microcystin Levels 2017-2025



Short term measures also addressed the City's response to finished water microcystin levels above the 0.3 µg/L. regarding public messaging and agency coordination. These measures included identifying specific agencies and principal contacts involved in decision making and communications and resources immediately available, such as alternate potable water.

2.8.2 Long Term Measures

Long-term measures include extending the City's shallow water intake. In September 2019, the City's Engineering Consultant submitted a report titled*; Conceptual Design of the Extension of Intake No. 2*. The document updated key elements of the 2004 Conceptual Design which comprised of extending Intake #2 (3,400 feet in length) to a depth of approximately 60 feet. The report included cost estimations, a preliminary schedule, water quality assessment, hydraulic analysis, pipe design and construction and permitting strategy. Deliverables initiated or completed in 2024 included deployment of water quality monitoring buoys, water quality modeling and hydraulic modeling. The City's consultant also completed *Contract Documents and Specifications* and *Engineering Drawings* (90% Design Submission) for review. This submission proposed extending the Intake approximately 4,850 feet to a water depth of approximately 80 feet.

Additional measures include continued and advanced source water protection activities. Source water protection activities are also part of the HAB Action Plan that the New York State Department of Environmental Conservation (NYSDEC) is developing in collaboration with steering committees.

3. CHEMICAL SAMPLING AND ANALYSIS

3.1 Organic and Inorganic Sampling Results

Skaneateles Lake water was sampled and analyzed for several chemicals and/or compounds during 2025. The analyses included: trihalomethanes (THM), haloacetic acids (HAA5), volatile organic compounds (VOC) with methyl-tertiary butyl ether (MTBE), synthetic organic compounds (SOC), inorganic chemicals and physical characteristics.

The monitoring results are listed in Tables Nos. 19-27, respectively. The frequency of sampling and methods of analyses were in accordance with 10 NYCRR Part 5-1 and/or Environmental Protection Agency (EPA) regulations. The results of the chemical analyses show that all concentrations were below the New York State Department of Health (NYS DOH) or EPA Maximum Contaminant Levels (MCLs).

Table 19 Total Trihalomethane Analyses of Chlorinated Skaneateles Lake Water for 2025

Results in ug/L	4000 East Genesee St.	Fuel Hub Truck Stop	Functional Comm. Corp.	McChesney Center	Quarterly Average (All Sites)
February 27	25.70	22.00	22.80	20.30	22.70
May 20	28.00	24.80	29.60	24.50	26.73
August 27	39.70	49.60	38.60	39.60	41.88
December 10	22.54	17.76	25.40	18.78	21.12
Locational Annual Average	28.99	28.54	29.10	25.77	

40/30 Certification Exemption per Disinfection Byproducts Rules: <40 ug/L

Table 20 Haloacetic Acids (5) Analyses of Skaneateles Lake Water for 2025

Results in ug/L	400 East Genesee St.	Fuel Hub Truck Stop	Functional Comm. Corp.	McChesney Center	Quarterly Average (All Sites)
February 27	17.70	15.70	17.40	15.00	16.45
May 20	22.20	20.70	19.90	20.70	20.88
August 27	20.80	18.10	20.40	16.90	19.05
December 10	17.20	16.90	19.20	18.80	18.03
Locational Annual Average	19.48	17.85	19.25	17.88	

Table 21 Volatile Organic Compounds, Vinyl Chloride and MTBE: February 27, 2025, Skaneateles Lake Water

Volatile Organic Compound	ug/L	Volatile Organic Compound	ug/L
Benzene	<0.5	Cis-1,3-Dichloropropene	<0.5
Bromobenzene	<0.5	trans-1,3-Dichloropropene	<0.5
Bromochloromethane	<0.5	Ethylbenzene	<0.5
Bromomethane	<0.5	Hexachlorobutadiene	<0.5
N-Butylbenzene	<0.5	Isopropylbenzene	<0.5
sec-Butylbenzene	<0.5	4-Isopropyltoluene	<0.5
tert-Butylbenzene	<0.5	Methylene Chloride	<0.5
Carbon Tetrachloride	<0.5	n-Propylbenzene	<0.5
Chlorobenzene	<0.5	Styrene	<0.5
Chloroethane	<0.5	1,1,1,2-Tetrachloroethane	<0.5
Chloromethane	<0.5	1,1,2,2-Tetrachloroethane	<0.5
2-Chlorotoluene	<0.5	Tetrachloroethene	<0.5
4-Chlorotoluene	<0.5	Toluene	<0.5
Dibromomethane	<0.5	1,2,3-Trichlorobenzene	<0.5
1,2-Dichlorobenzene	<0.5	1,2,4-Trichlorobenzene	<0.5
1,3-Dichlorobenzene	<0.5	1,1,1-Trichloroethane	<0.5
1,4-Dichlorobenzene	<0.5	1,1,2-Trichloroethane	<0.5
Dichlorodifluoromethane	<0.5	Trichloroethene	<0.5
1,1-Dichloroethane	<0.5	Trichlorofluoromethane	<0.5
1,2-Dichloroethane	<0.5	1,2,3-Trichloropropane	<0.5
1,1-Dichloroethene	<0.5	1,2,4-Trimethylbenzene	<0.5
cis-1,2-Dichloroethene	<0.5	1,3,5-Trimethylbenzene	<0.5
trans-1,2-Dichloroethene	<0.5	m-Xylene	<0.5
1,2-Dichloropropane	<0.5	o-Xylene	<0.5
1,3-Dichloropropane	<0.5	p-Xylene	<0.5
2,2-Dichloropropane	<0.5	Vinyl Chloride	<0.5
1,1-Dichloropropene	<0.5	MTBE	<0.5

Table 22 Volatile Organic Compounds, Vinyl Chloride and MTBE: May 20, 2025, Skaneateles Lake Water

Volatile Organic Compound	ug/L	Volatile Organic Compound	ug/L
Benzene	<0.5	Cis-1,3-Dichloropropene	<0.5
Bromobenzene	<0.5	trans-1,3-Dichloropropene	<0.5
Bromochloromethane	<0.5	Ethylbenzene	<0.5
Bromomethane	<0.5	Hexachlorobutadiene	<0.5
N-Butylbenzene	<0.5	Isopropylbenzene	<0.5
sec-Butylbenzene	<0.5	4-Isopropyltoluene	<0.5
tert-Butylbenzene	<0.5	Methylene Chloride	<0.5
Carbon Tetrachloride	<0.5	n-Propylbenzene	<0.5
Chlorobenzene	<0.5	Styrene	<0.5
Chloroethane	<0.5	1,1,1,2-Tetrachloroethane	<0.5
Chloromethane	<0.5	1,1,2,2-Tetrachloroethane	<0.5
2-Chlorotoluene	<0.5	Tetrachloroethene	<0.5
4-Chlorotoluene	<0.5	Toluene	<0.5
Dibromomethane	<0.5	1,2,3-Trichlorobenzene	<0.5
1,2-Dichlorobenzene	<0.5	1,2,4-Trichlorobenzene	<0.5
1,3-Dichlorobenzene	<0.5	1,1,1-Trichloroethane	<0.5
1,4-Dichlorobenzene	<0.5	1,1,2-Trichloroethane	<0.5
Dichlorodifluoromethane	<0.5	Trichloroethene	<0.5
1,1-Dichloroethane	<0.5	Trichlorofluoromethane	<0.5
1,2-Dichloroethane	<0.5	1,2,3-Trichloropropane	<0.5
1,1-Dichloroethene	<0.5	1,2,4-Trimethylbenzene	<0.5
cis-1,2-Dichloroethene	<0.5	1,3,5-Trimethylbenzene	<0.5
trans-1,2-Dichloroethene	<0.5	m-Xylene	<0.5
1,2-Dichloropropane	<0.5	o-Xylene	<0.5
1,3-Dichloropropane	<0.5	p-Xylene	<0.5
2,2-Dichloropropane	<0.5	Vinyl Chloride	<0.5
1,1-Dichloropropene	<0.5	MTBE	<0.5

Table 23 Volatile Organic Compounds, Vinyl Chloride and MTBE: August 27, 2025, Skaneateles Lake Water

Volatile Organic Compound	ug/L	Volatile Organic Compound	ug/L
Benzene	<0.5	Cis-1,3-Dichloropropene	<0.5
Bromobenzene	<0.5	trans-1,3-Dichloropropene	<0.5
Bromochloromethane	<0.5	Ethylbenzene	<0.5
Bromomethane	<0.5	Hexachlorobutadiene	<0.5
N-Butylbenzene	<0.5	Isopropylbenzene	<0.5
sec-Butylbenzene	<0.5	4-Isopropyltoluene	<0.5
tert-Butylbenzene	<0.5	Methylene Chloride	<0.5
Carbon Tetrachloride	<0.5	n-Propylbenzene	<0.5
Chlorobenzene	<0.5	Styrene	<0.5
Chloroethane	<0.5	1,1,1,2-Tetrachloroethane	<0.5
Chloromethane	<0.5	1,1,2,2-Tetrachloroethane	<0.5
2-Chlorotoluene	<0.5	Tetrachloroethene	<0.5
4-Chlorotoluene	<0.5	Toluene	<0.5
Dibromomethane	<0.5	1,2,3-Trichlorobenzene	<0.5
1,2-Dichlorobenzene	<0.5	1,2,4-Trichlorobenzene	<0.5
1,3-Dichlorobenzene	<0.5	1,1,1-Trichloroethane	<0.5
1,4-Dichlorobenzene	<0.5	1,1,2-Trichloroethane	<0.5
Dichlorodifluoromethane	<0.5	Trichloroethene	<0.5
1,1-Dichloroethane	<0.5	Trichlorofluoromethane	<0.5
1,2-Dichloroethane	<0.5	1,2,3-Trichloropropane	<0.5
1,1-Dichloroethene	<0.5	1,2,4-Trimethylbenzene	<0.5
cis-1,2-Dichloroethene	<0.5	1,3,5-Trimethylbenzene	<0.5
trans-1,2-Dichloroethene	<0.5	m-Xylene	<0.5
1,2-Dichloropropane	<0.5	o-Xylene	<0.5
1,3-Dichloropropane	<0.5	p-Xylene	<0.5
2,2-Dichloropropane	<0.5	Vinyl Chloride	<0.5
1,1-Dichloropropene	<0.5	MTBE	<0.5

Table 24 Volatile Organic Compounds, Vinyl Chloride and MTBE: December 10, 2025, Skaneateles Lake Water

Volatile Organic Compound	ug/L	Volatile Organic Compound	ug/L
Benzene	<0.5	Cis-1,3-Dichloropropene	<0.5
Bromobenzene	<0.5	trans-1,3-Dichloropropene	<0.5
Bromochloromethane	<0.5	Ethylbenzene	<0.5
Bromomethane	<0.5	Hexachlorobutadiene	<0.5
N-Butylbenzene	<0.5	Isopropylbenzene	<0.5
sec-Butylbenzene	<0.5	4-Isopropyltoluene	<0.5
tert-Butylbenzene	<0.5	Methylene Chloride	<0.5
Carbon Tetrachloride	<0.5	n-Propylbenzene	<0.5
Chlorobenzene	<0.5	Styrene	<0.5
Chloroethane	<0.5	1,1,1,2-Tetrachloroethane	<0.5
Chloromethane	<0.5	1,1,2,2-Tetrachloroethane	<0.5
2-Chlorotoluene	<0.5	Tetrachloroethene	<0.5
4-Chlorotoluene	<0.5	Toluene	<0.5
Dibromomethane	<0.5	1,2,3-Trichlorobenzene	<0.5
1,2-Dichlorobenzene	<0.5	1,2,4-Trichlorobenzene	<0.5
1,3-Dichlorobenzene	<0.5	1,1,1-Trichloroethane	<0.5
1,4-Dichlorobenzene	<0.5	1,1,2-Trichloroethane	<0.5
Dichlorodifluoromethane	<0.5	Trichloroethene	<0.5
1,1-Dichloroethane	<0.5	Trichlorofluoromethane	<0.5
1,2-Dichloroethane	<0.5	1,2,3-Trichloropropane	<0.5
1,1-Dichloroethene	<0.5	1,2,4-Trimethylbenzene	<0.5
cis-1,2-Dichloroethene	<0.5	1,3,5-Trimethylbenzene	<0.5
trans-1,2-Dichloroethene	<0.5	m-Xylene	<0.5
1,2-Dichloropropane	<0.5	o-Xylene	<0.5
1,3-Dichloropropane	<0.5	p-Xylene	<0.5
2,2-Dichloropropane	<0.5	Vinyl Chloride	<0.5
1,1-Dichloropropene	<0.5	MTBE	<0.5

Table 25 Synthetic Organic Compound Analyses of Skaneateles Lake Water May 20, 2025

Part 5 Group Number	Parameter	EPA Standard in ug/L	Results in ug/L
1	Alachlor	0.2	Not Detected
1	Aldicarb	3	Not Detected
1	Aldicarb sulfone	2	Not Detected
1	Aldicarb sulfoxide	4	Not Detected
1	Atrazine	3	Not Detected
2	Benzo(a)pyrene	0.2	Not Detected
1	Carbofuran	40	Not Detected
1	Chlordane, Total	2	Not Detected
2	Bis(2-ethylhexyl)pthalate	6	Not Detected
1	1,2-dibromo-3-chlorpropane	0.2	Not Detected
1	2,4-D	50	Not Detected
2	Dinoseb	7	Not Detected
1	Endrin	2	Not Detected
1	1,2-dibromoethane (EDB)	0.05	Not Detected
1	Heptachlor	0.4	Not Detected
1	Heptachlor expoxide	0.2	Not Detected
2	Hexachlorobenzene	1	Not Detected
1	Gamma-BHC (Lindane)	0.2	Not Detected
+1	Methoxychlor	40	Not Detected
1	Pentachlorophenol	1	Not Detected
1	PCB, Total	0.5	Not Detected
2	Picloram	500	Not Detected
2	Simazine	4	Not Detected
1	Toxaphene	3	Not Detected
1	2,4,5-TP Silvex	10	Not Detected
2	3-Hydroxy Carbofuran	5	Not Detected
2	Aldrin	5	Not Detected
2	Butachlor	50	Not Detected
2	Carbaryl	5	Not Detected
2	Dalapon	50	Not Detected
2	Dicamba	50	Not Detected
2	Dieldrin	5	Not Detected
2	Glyphosate	50	Not Detected
2	Hexachlorocyclopentadiene	5	Not Detected
2	Methomyl	5	Not Detected
2	Metolachlor	50	Not Detected

2	Metribuzin	50	Not Detected
2	Oxamyl	5	Not Detected
2	Propachlor	50	Not Detected

Table 26 Inorganic Chemicals and Physical Characteristics Analysis of Skaneateles Lake Water May 20, 2025

Parameter	EPA Standard in mg/L	Results in mg/L
Antimony	0.006	Not Detected
Arsenic	0.010	Not Detected
Barium	2.0	0.0233
Berylium	0.004	Not Detected
Bromate	0.01	Not Detected
Cadmium	0.005	Not Detected
Chlorite	1.0	Not Detected
Chromium	0.10	Not Detected
Cyanide	0.02	Not Detected
Iron	0.30	Not Detected
Lead	0.015 *	Not Detected
Mercury	0.002	Not Detected
Nickel	None	Not Detected
Selenium	0.05	Not Detected
Thallium	0.002	Not Detected
Fluoride	2.2	0.52
Chloride	250	22.8 **
Magnesium	None	7.19
Manganese	0.30	Not Detected
Potassium	None	1.10
Silver	0.10	Not Detected
Sodium	None	11.4
Sulfate	250	10.6 **
Color	15 units	<5
Odor	3 units	1 unit
Nitrate	10.0	0.27 **
Nitrite	1.0	Not Detected
Zinc	5.0	Not Detected

*EPA Action Level **August 27, 2025

Table 27 Radionuclide Analyses of Skaneateles Lake Water August 9, 2023

Parameter	EPA Standard in picocuries/liter (pCi/l)	Results in pCi/l
Gross Alpha Particle	15	0.928
Radium 226	5	0.487
Radium 228	5	0.337

Table 28 Emerging Contaminants Detected – PFAS / 1-4 Dioxane Analysis of Skaneateles Lake Water

Contaminant	Unit	Regulatory Limit (MCL)	MCLG	Level Detected	Sample Date	Violation
Perfluorooctanoic acid (PFOA) (1)	ng/L	10	N/A	0.96 (2)	11/15/2022	No
Perflurobutanoic acid (PFBA) (10)	ng/L	50,000	N/A	1.9	11/27/2023	No

Notes:
1: Released into the environment from widespread use in commercial and industrial applications
2: Level detected below reporting limit of 1.8 ng/L

4. SKANEATELES LAKE WATERSHED PROTECTION PROGRAM

4.1 Watershed Inspection Program Overview

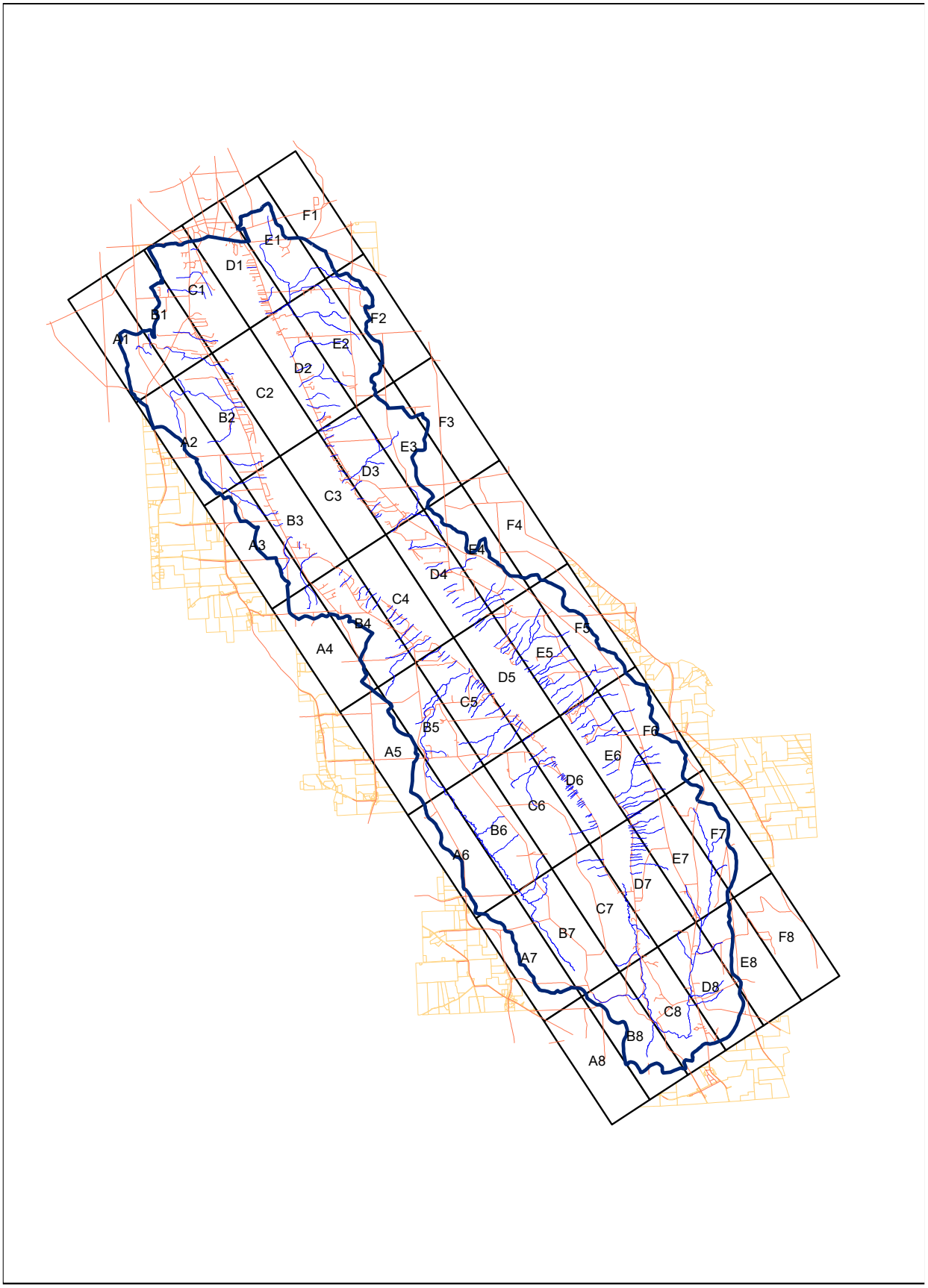
The Watershed Protection Program operated by the City of Syracuse consisted of (a) an Inspection Program to detect violations of the Watershed Rules and Regulations, NYSDOH and county Sanitary Codes, NYSDEC Environmental Conservation Law (ECL), Navigation Law and local rules and regulations, (b) direct involvement in the procedures for installation of new and repair of existing OWTS, (c) reviewing all design and site plans for proposed building and land disturbing activity and (d) the OWTS Inspection Program.

Two City of Syracuse personnel routinely patrol the lake and watershed: a Watershed Inspector and a Water Department Sanitarian. Typically, there is an increase in regulated activities within the watershed during the summer months (May-September). This is due, in large part, to the influx of seasonal residents and non-resident recreational users (boaters, anglers, hikers, etc.). The activities include construction and/or repair of dwellings, OWTS and shoreline structures and recreation.

To address this increased seasonal activity, Inspectors conduct intensive morning investigations of lakeshore properties. This includes walking properties, looking under structures, inspecting OWTS failures, and investigating construction activities of any kind. When required, a boat detail is implemented for those areas that are difficult to access by truck or foot. To more effectively survey the entire land area within the watershed annually, watershed zones were digitally overlaid and labeled on a GIS map in 2012 (Figure 10). The zones appear on the Skaneateles Lake Watershed Map as 48 rectangular grids, each representing approximately 870 acres of land. Inspectors are assigned to individual zones based on the time of year. Zones comprising of lakefront seasonal cottages are inspected during the summer months and remote areas located off seasonal roads are typically inspected in the spring and fall.

The afternoon schedule allows for flexibility, and no specific areas are selected for inspection. This affords the Inspectors the latitude to conduct general inspections, pursue compliance of outstanding violations, conduct site surveys for OWTS proposals and meet with property owners, Environmental Conservation Officers (ECOs), Building Inspectors, or County Public Health Technicians/Sanitarians.

Figure 10 Skaneateles Lake Watershed Inspection Zones



4.2 Onsite Wastewater Treatment System Review and Inspection

Soil tests were witnessed by the Inspectors on 24 occasions in 2025. All soil tests were done in accordance with 10 NYCRR Part 75 (Appendix 75-A) and witnessed by the inspection team. All proposed systems for new dwellings require a plan designed by a licensed Design Professional. Typically, property owners hire a private consultant. However, Cortland County Department of Health personnel are responsible for conventional system design in their county. Sites requiring non-conventional systems require designs by a private consultant. Plans are submitted to the Department of Water for review and comment, and to the respective county health department for approval or rejection based on 10 NYCRR Part 75 guidelines. A total of 23 OWTS design proposals for new construction or alternative engineering design were recommended for approval. No additional proposals were reviewed for repairing existing septic system components. Watershed personnel also conducted backfill inspections on repairs, as well as assisted county Sanitarians and design engineers with final inspections for new construction.

4.3 Housing Starts in the Watershed

A total of 19 building permits were issued for new dwellings during 2025 (Table No. 29 & Fig. 11). Eighteen of the permits were in Onondaga County. Seven were in the town of Skaneateles (four lakefront), two were in the Village of Skaneateles (no lakefront) and nine were issued in the town of Spafford (four lakefront). One permit was issued in Cayuga County, (lakefront in the town of Sempronius).

Table 29 Construction Activity in the Skaneateles Watershed for 2025

Township	New Construction	Lakefront*	Additions & Renovations	Lakefront**
Village of Skaneateles	2	0	1	0
Town of Skaneateles	7	4	7	3
Spafford	9	4	6	4
Niles	0	0	4	4
Sempronius	1	1	0	0
Scott	0	0	0	0
Total	**19**	**9**	**18**	**11**

* Included in new construction figures

** Included in additions & renovations figures

4.4 Building Permit Application Review

The Watershed Protection Program reviewed a total of 78 Building and Zoning Permit Applications in 2025. As discussed above, 19 were new construction. Eighteen applications were additions or renovations to existing dwellings, and the remaining 41 applications included shoreline structures, accessory structures, landscape features or proposed subdivisions and lot line relocations. Figures 11 through 14 illustrate new construction and renovation proposals reviewed by the Watershed Protection Program and monitored throughout project duration in the Skaneateles Lake Watershed since 1993.

Figure 11 New Construction Activity within the Skaneateles Lake Watershed 1993-2025

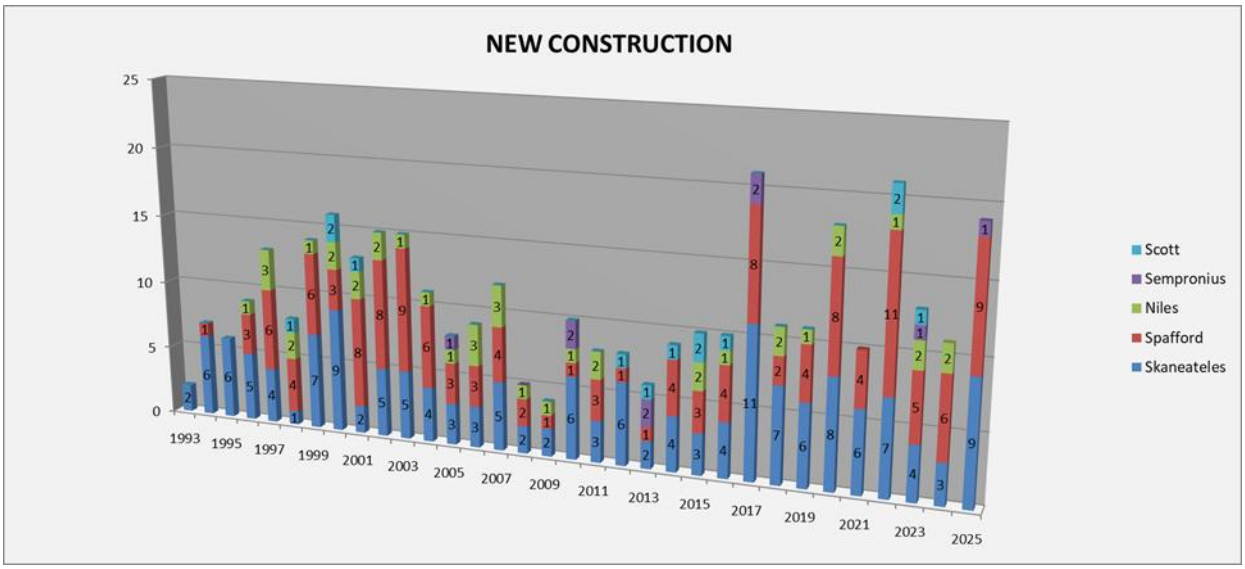


Figure 12 Renovation Activity within the Skaneateles Lake Watershed 1993-2025

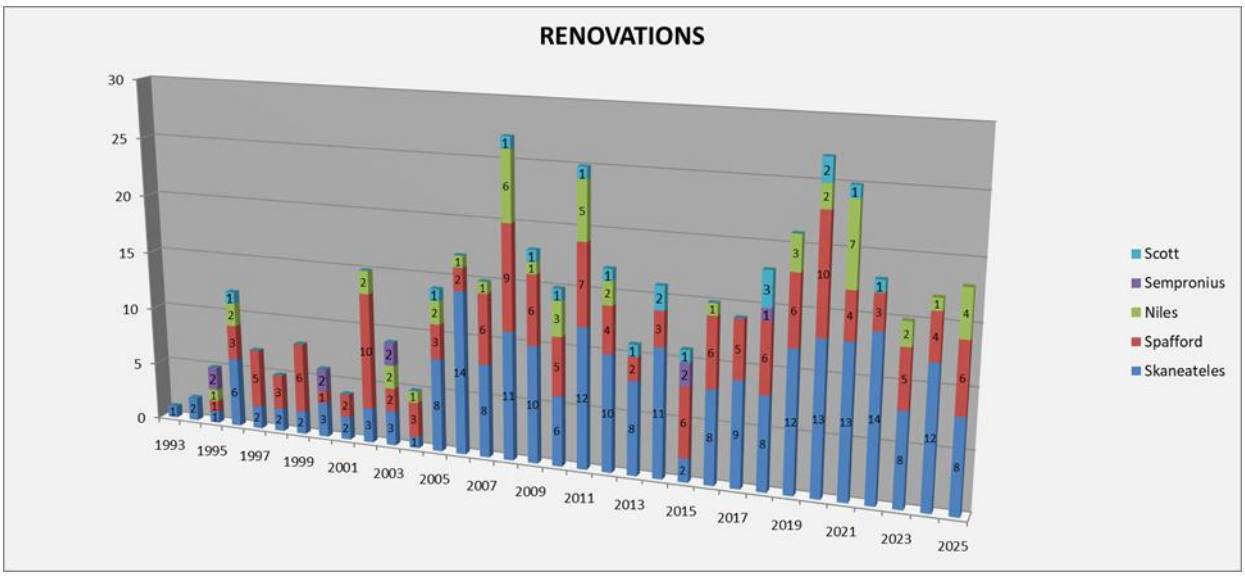


Figure 13 Map of New Construction Activity within the Skaneateles Lake Watershed 1993-2025

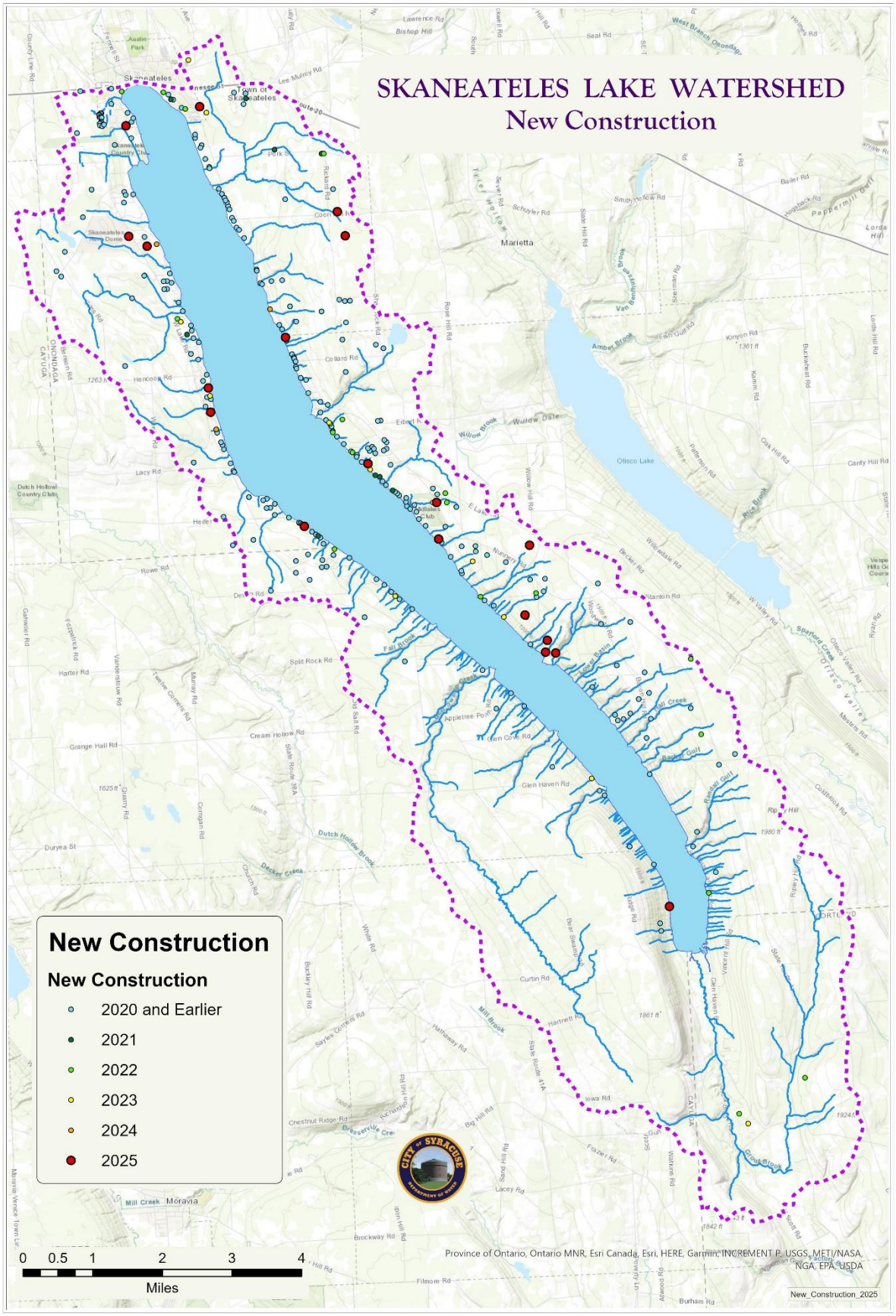
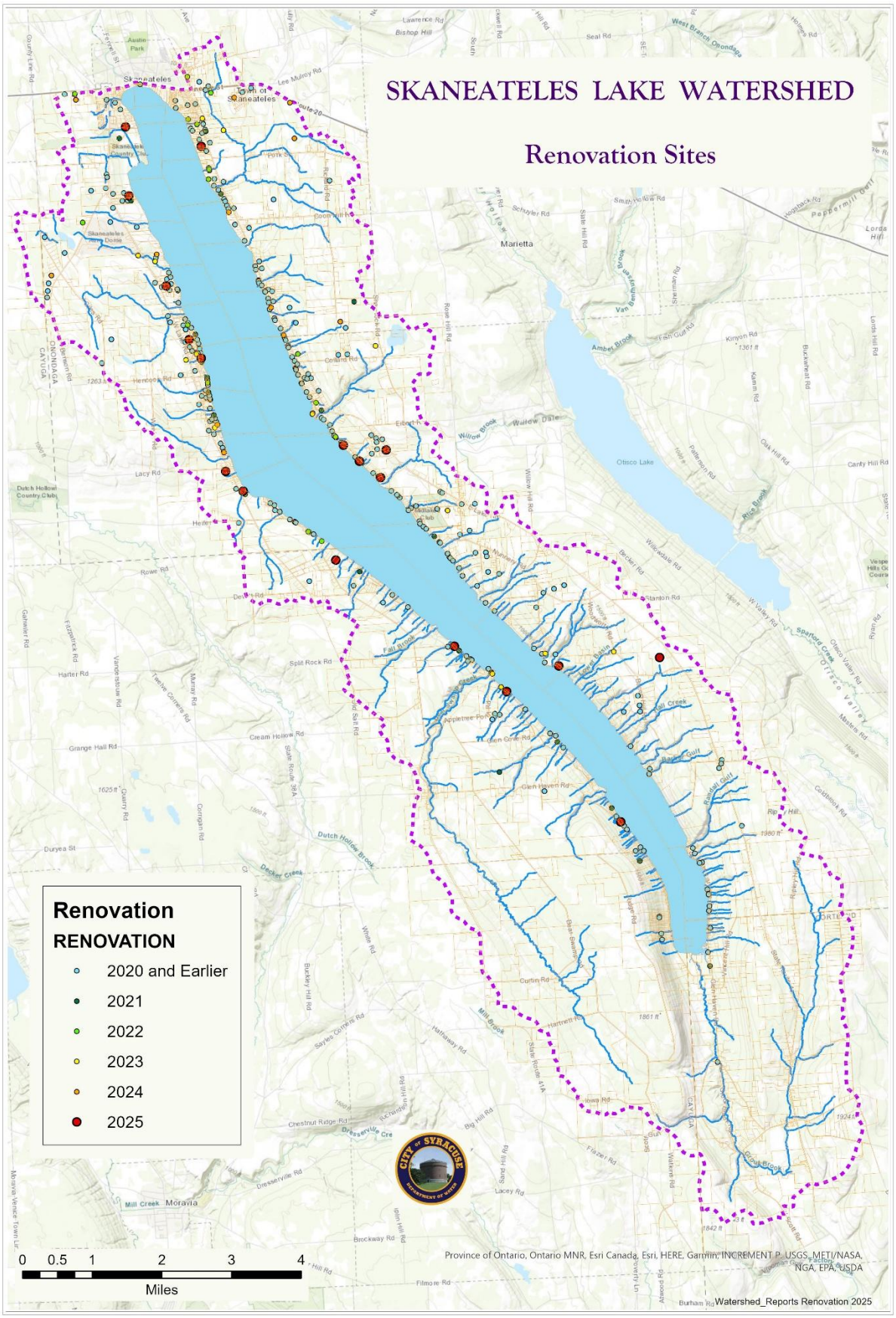


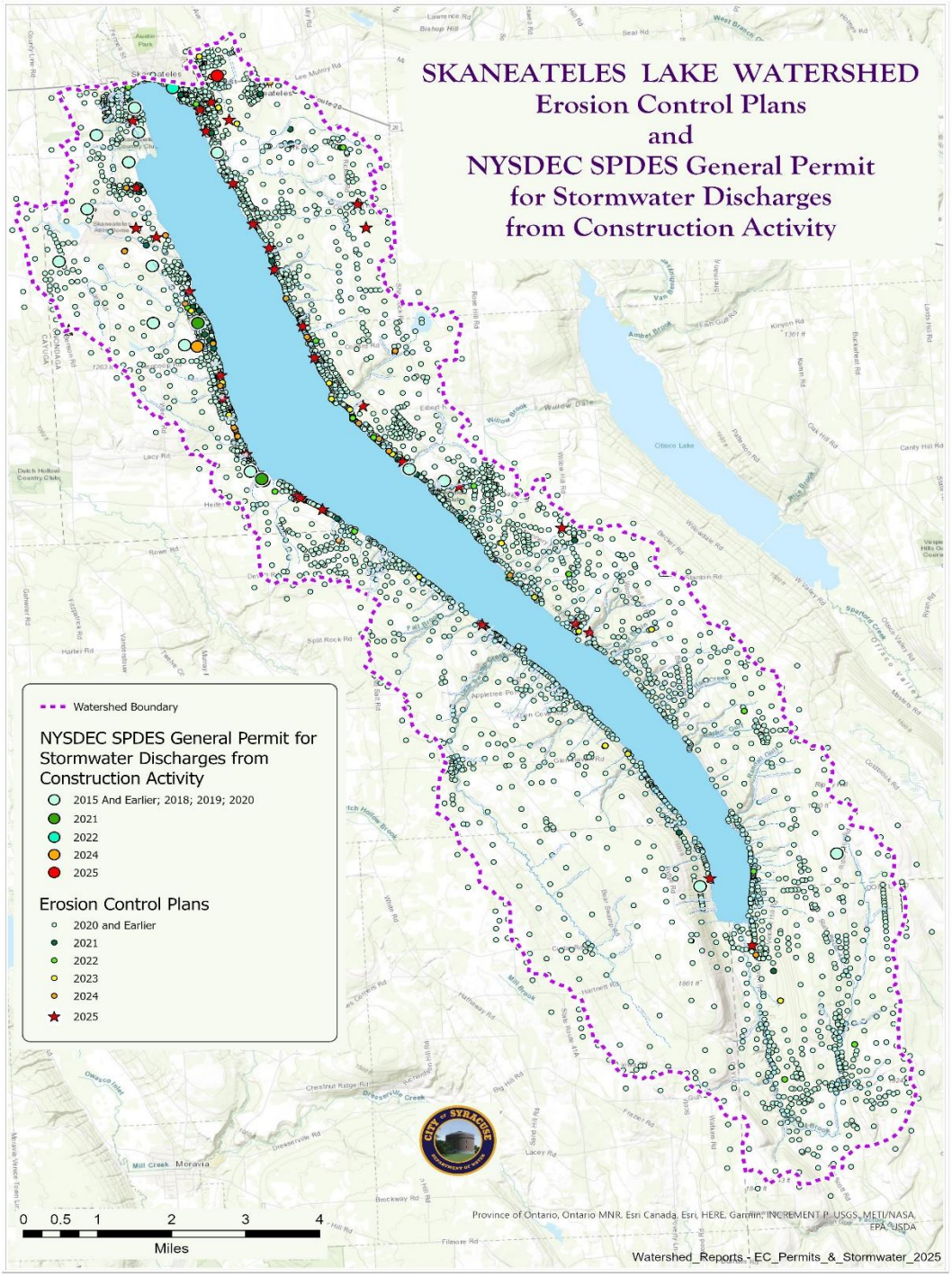
Figure 14 Map of Renovation Activity within the Skaneateles Lake Watershed 1993-2025



4.5 Erosion and Sediment Control Plan Review

As of 2004, the Skaneateles Watershed Rules and Regulations require property owners disturbing 5,000 square feet or greater of land in defined environmentally sensitive areas to submit a Sediment and Erosion Control Plan (SECP) to the City of Syracuse for review. In 2025, 35 SECPs were reviewed. A NYSDEC SPDES General Permit for Stormwater Discharges (GP-0-25-001) is required in New York State for all construction activity over one acre. One permit was issued for construction activity in the Watershed in 2025 under GP-0-25-001. Figure 15 illustrates SECP's review and SPDES General Permits issued within the Skaneateles Lake Watershed since 2004.

Figure 15 Erosion Control Plans/NYSDEC SPDES Permits



4.6 Skaneateles Lake Watershed Rules and Regulations Violations

A total of 11 violations of the Skaneateles Lake Watershed Rules and Regulations were recorded in 2025. The violations are discussed in detail below.

4.6.1 County Sanitary Code Violations

Ten violations of County Sanitary Code were reported in 2025. Nine violations were the result of failure of property owners to renew operation and maintenance (O&M) contracts for enhanced treatment units (ETUs) or aerobic treatment units (ATUs). One violation notice was issued for modifications to existing septic system components without County Health Department approval (Table No. 28 at end of next section). As a result of Syracuse Water Department violation notices issued to property owners, nine O&M contracts were renewed with service contractors, and one as-built septic system repair plan was recorded.

The enforcement of Sanitary Code violations is the responsibility of the respective county health departments. Alleged violators are issued a Violation Notice from the Watershed Protection Program and given five working days to reply and declare their intent to affect the necessary repairs. If there is no response, or if the property owner refuses to abate the problem, the violation is forwarded to the respective county health department for enforcement. Sanitary Code violations identified by Watershed Inspectors since 1994 are shown in Figure 16.

Figure 16 Skaneateles Lake Watershed Sanitary Code Violations 1994-2025

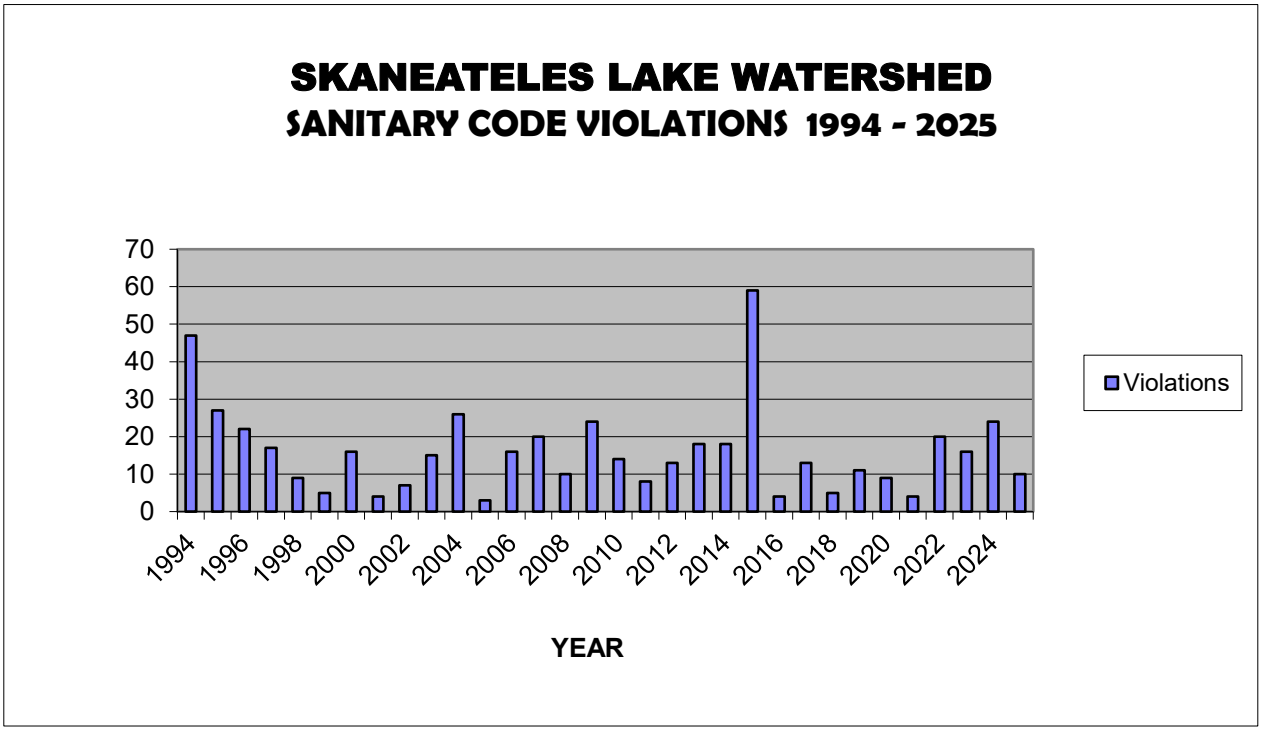
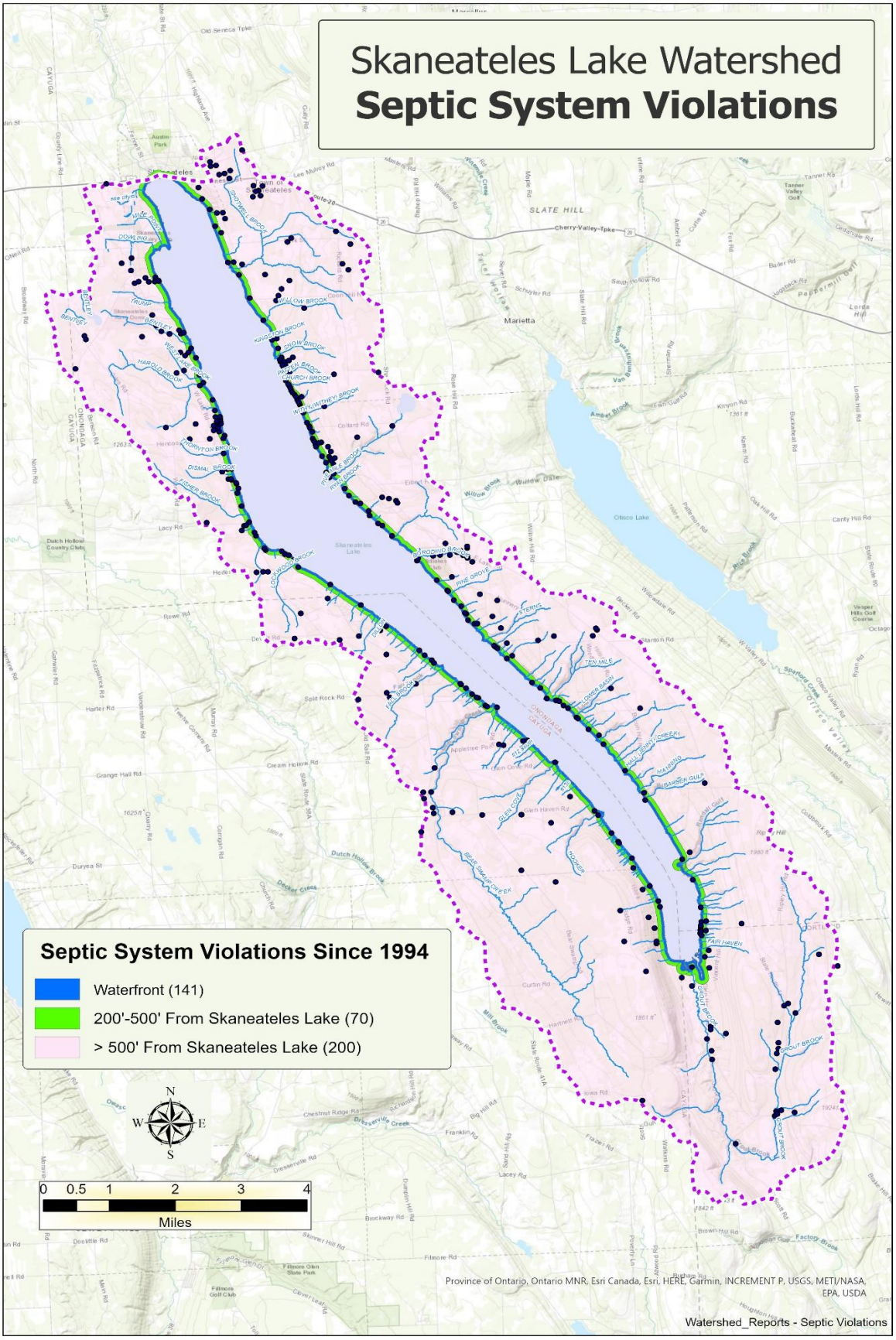


Figure 17 Sanitary Code Violation – Proximity To Skaneateles Lake 1994-2025



4.6.2 Erosion and Sediment Control Violations

Twelve violations were issued for non-compliance of sediment and erosion control practices in 2025. All violations were abated in 2025 (Table No. 30).

Table 30 Violations for the Past Five Years: NYSDOH and Local DOH Sanitary Code Erosion and Sediment Control

Year	Number or Violations DOH/Sanitary Code	Number of Violations Erosion and Sediment	Abated or Pending Action by City	Reported to DOH or NYSDEC
2021	4	9	13	0
2022	20	13	33	1
2023	16	13	29	0
2024	24	18	42	0
2025	10	12	22	0

* New regulations became effective in 2004

4.6.3 Petroleum/Hazardous Material Spills

There were four potentially hazardous incidents or material releases identified by city personnel or reported by NYSDEC Division of Environmental Remediation in 2025.

- **Wednesday, March 19, 2025** – NYSDEC Spill Hotline received a report of a granular fertilizer release onto Rt. 41A in the Town of Niles. City of Syracuse Water Dept. personnel were dispatched to the spill location and observed the clean-up effort by the farm operator and Town of Niles DPW.

- **Monday, May 5, 2025** – A farm operator in the Town of Spafford reported a silage leachate discharge to an alfalfa field, originating from a storage pond underdrain leak. Multi-agency coordination—involving the Watershed Protection Program, Skaneateles Lake Watershed Agricultural Program (SLWAP), and NYSDEC Water Quality Section assisted in containment and remediation efforts. Leachate runoff from the field resulted in observations of foaming and water discoloration in the upper sections of a tributary to Lower Basin Brook. Water quality showed significant improvement at the watercourse outlet due to natural aeration, mixing, and dilution.
  Containment and remediation actions included:
  - Leak Sealing: The 4” drain outlet was successfully plugged and capped by the farm operator
  - Effluent Diversion: leachate flowing towards Lower Basin Brook was diverted using an earthen berm into a nearby alfalfa field to prevent further stream contamination
  - Pond Drawdown: With the assistance of the SLWAP and an agricultural contractor, the farm operator used hydraulic and submersible pumps to remove leachate. Over 15 tanker trucks transported the material to local manure lagoon
  - Several tears were identified in the pond liner and repaired following complete dewatering

- **Tuesday, July 29, 2025 –** A release of liquid manure was reported on an agricultural field in the town of Skaneateles due to a manure dragline system failure. The failure resulted in a significant amount of ponded manure accumulating near a road ditch. A minimal amount of manure was observed within the ditch.

Remediation and Response Actions

- Immediate containment and cleanup efforts were initiated following the discovery of the release. The farm operator constructed an earthen berm above the road ditch to prevent further movement of the liquid manure
- Simultaneously, the manure dragline contractor began spreading and incorporating liquid manure into the surrounding fields to facilitate absorption
- The remediation process also included scraping all liquid manure out of the affected 20-foot section of the road ditch, which was then land-applied to the field
- The contractor completed the spreading and incorporation of the remaining manure-saturated ground that was contained around the hose area

- **Friday, November 14, 2025 -** A NYSDEC Regional Spill Engineer contacted this office to report a motor vehicle accident resulting in diesel discharging to a road ditch which conveys stormwater to Grout Brook. Absorbent booms were placed across the ditch in several locations downstream of the spill location. A NYSDEC Spill contractor was dispatched to perform clean-up which included mobilizing a vac truck to skim petroleum sheens collecting upstream of the absorbent booms.

4.7 Composting Toilet Operation

Currently 35 property owners utilize composting toilets provided by the Syracuse Water Department. Compost toilet users are responsible for removing finished compost from their units and emptying the compost in clean 5-gallon buckets provided by the Syracuse Water Department. Finished compost is collected in the spring by City Water Department personnel and placed in 4-foot (w) x 4-foot (l) x 4 foot (d) wooden containers, allowing for additional composting on the city's Glen Haven property. The storage containers have been constructed to allow thermophilic composting (internal temperature exceeding 105 degrees Fahrenheit) to take place, further reducing or eliminating any pathogens remaining from the collected compost. Compost piles are monitored to ensure the required temperatures are attained. Carbon based materials such as grass clippings, mulch and wood chips are added to the piles periodically to maintain a balanced carbon/nitrogen ratio. A balanced ratio allows for optimum digestion of compost by microorganisms, resulting in accelerated temperatures in the pile. Sampling for fecal coliform coincides with monitoring thermophilic conditions within the compost pile to ensure pathogen reduction. Compost is typically stored for a minimum of three years allowing for a significant reduction in volume and pathogens before the remaining organic material is mixed within the leaf litter.

5. FILTRATION AVOIDANCE WAIVER

5.1 Filtration Avoidance Waiver Conditions

The City of Syracuse applied for and received a filtration avoidance waiver extension on June 28, 2004. The waiver has no termination date and will remain in effect for as long as the City complies with the conditions of that filtration avoidance waiver.

Progress continues to be made on the programs implemented by the City in order to enhance the Skaneateles Lake Watershed Management Program. The programs are: The Data Gathering and Management Program, Conservation Easement Acquisition Program (now complete), the Skaneateles Lake Watershed Agricultural Program (SLWAP), and the Water Quality Public Education Program.

For a detailed discussion of these programs, refer to the Skaneateles Lake Watershed Program Annual Report 2025-2026.

6. PESTICIDE AND FERTILIZER INVENTORY

6.1 Agricultural Pesticide and Fertilizer Use Survey

The annual Agricultural Survey was conducted by the Skaneateles Lake Watershed Agricultural Program staff. A total of 34 farm operators were involved in the survey. These farms are actively participating in the Skaneateles Lake Watershed Agricultural Program (SLWAP) with the majority of farms located entirely in the watershed. A total of 29,936 acres were found to have been worked by the 34 farms with active Whole Farms Plans.

A total of 1,025 tons of granular fertilizer and 23,636 gallons of liquid fertilizer were applied to cropland. A total of 637 tons of lime were applied to cropland. A total of 3,242 tons of manure, and 21,509,842 gallons were utilized for nutrient value and soil organic matter enhancement. This manure was produced by approximately 2,059 animals (1,567 dairy animals, 76 horses, 102 sheep, 90 beef, 170 bison, 15 alpacas, 37 goats, 2 pigs).

A total of 4,495 gallons of liquid pesticide and 400 pounds of granular pesticide were applied to cropland. These numbers represent totals for all pesticides (herbicides, insecticides, and fungicides) that were applied at the rate specified by the label.

A total of 128,386 gallons of diesel and 18,358 gallons of gasoline were used by the 34 farms.

Watershed Inspectors conducted an annual survey of agricultural operations focusing on petroleum and chemical storage, disposal practices and solid and hazardous waste. There was no evidence of petroleum spills or leakage from bulk storage tanks, dispensers, or hoses during the inspections.

6.2 Pesticide Use by State and Local Departments of Transportation

Herbicides were not used in the Skaneateles Lake Watershed by Onondaga, Cayuga or Cortland County DOTs in 2025. The NYSDOT Onondaga Residency and Cortland Residency applied herbicides along Route 41 and Rt. 41A (Onondaga Residency) to manage roadside vegetation and for invasive species control. Pesticides and herbicides applied included Arsenal Powerline, Plainview SC, RoundupPro Concentrate, Milestone, Plateau, Escort XP, Arsenal Applicators Concentrate, Method 240 SL and Esplanade 200 SC. The NYSDOT Cayuga County Residency did not apply pesticides in the Skaneateles Lake Watershed. This information was based on personal communication and/or information provided by Regional Engineers for the respective NYS Residences, the Environmental Specialist with the Onondaga County DOT, General Foreman of the Cayuga County DOT and the Superintendent of Highways for the Cortland County DOT.

7. PETROLEUM AND HAZARDOUS MATERIALS INVENTORY

7.1 Inventory

Petroleum products were the most abundant and potentially hazardous materials stored on the watershed. In addition to the petroleum products stored on farms (see above), 18 commercial

enterprises and four municipal facilities stored a total of 51,505 gallons of petroleum products as of January 1, 2026. Gasoline accounted for 34,350 gallons of the total. Other products stored were propane (47,590 gallons), diesel (12,325 gallons), heating oil (1,100 gallons) and used oil (3,730 gallons).

Road salts are used by the local highway departments and the NYSDOT for vehicle safety. There is one covered storage facility in the Town of Scott. The average rate of application of salt or salt/sand mixtures on watershed roads by each agency was unchanged from previous years.

7.2 Road Ditch Survey

All road ditches within the watershed are inspected routinely for contamination sources. Watershed Inspectors document evidence of sewage discharge, agricultural runoff, petroleum spills, etc., on Survey Sheets and take necessary steps to identify the source of contamination through sampling point source discharges and investigating property records.

8. WATERSHED MANAGEMENT PROJECTS

8.1 Watershed Management Approach to Controlling Hemlock Woolly Adelgid (HWA)

HWA was identified in the Skaneateles Lake Watershed in 2014. Once infested with HWA, mature hemlock trees die within four to 20 years. The hemlock loss and replacement with hardwood species has the potential to impact water quality by altering nutrient cycling in the watershed and changing water temperature and water quantity going into the lake over the course of the year. Hemlocks' deep shade and often streamside habitat helps keep streams cool, and their evergreen shade keeps snow on the ground into the spring, providing cold runoff into groundwater farther into the growing season. Because hemlocks draw the most water during spring and fall, and relatively little in the summer, they also help stabilize stream flows.

HWA has been found on both shores of Skaneateles Lake in the southern portion of the lake. As of February 2026, the northernmost points where HWA has been found are around Fire Lane 22A on the western shore (reported in 2018), and Red Tail Lane on the eastern shore (reported in 2024). (For the most up to date information, please visit the NY iMapInvasives map at nyimapinvasives.org/data-and-maps). To minimize the spread of HWA, the City of Syracuse Water Department collaborated with the Onondaga County Soil and Water District, Cornell University, CCE of Onondaga County and several volunteers residing with the watershed. In May 2015, 100 Eastern Hemlock trees were planted within this region of the watershed to grow populations of biological controls to resist the spread of HWA.

Three insects that feed on HWA (biocontrols) have been released in the Skaneateles Watershed in 2015 and 2016. These are a beetle referred to as 'Little Larry', Laricobius nigrinus, and two species of silver fly, Leucopis piniperda and L. argenticollis. All three species are imported from their native range in the Northwestern US where they are natural predators of HWA. Establishment has not been verified in the Skaneateles watershed for any of the three species, but establishment can take many years to be detected.

To enhance and support this biocontrol management option, in November of 2017 a new, $1.2 million biocontrol laboratory was established on the Cornell University campus, focusing on researching and rearing biological controls to stop the spread of HWA. The lab is funded through

NYSDEC's monies from the NYS Environmental Protection Fund and is headed by a Cornell entomologist.

Biological control is a long-term solution for HWA, but landowners with trees that are currently infested are strongly encouraged to consider treatment of their trees. Treatment is relatively inexpensive and lasts for three - seven years. More information on HWA management options can be found on the NYS Hemlock Initiative website (NYSHI), nyshemlockinitiative.info.

In 2018, the agencies included HWA-specific iMap-Invasives training in their HWA workshop. IMap-Invasives is an online, GIS-based data management system used to assist citizen scientists and natural resource professionals working to protect our natural resources from the threat of invasive species. Attendees learned how to identify and report HWA infestations and, of equal importance, the absence of infestations around the Skaneateles watershed. Since the training, workshop participants on their own have logged over three dozen entries in the database project shared between CCE of Onondaga and the NYSHI. Including first reports of known infestations in the watershed, this citizen science effort has largely contributed to the NYSHI research is proving to be an efficient use of agency resources and aids our partners at the state PRISMs in following early detection, rapid response protocol.

CCE of Onondaga and the NYSHI have continued to collaborate on educational events around HWA. Most recently, in 2025, CCE of Onondaga, the NYSHI and Skaneateles Lake Association partnered on two educational programs. The first was a 'lunch and learn' webinar on March 25th, 2025, with 38 attendees. During the webinar, the NYS Hemlock Initiative shared an update on HWA, as well as the changing landscape of HWA management options. The Skaneateles Lake Association also shared an update on their work to treat hemlock trees throughout the Skaneateles Lake Watershed. The webinar was recorded and is available on the CCE Onondaga YouTube page. The second event, held on October 21st, 2025, at Lourdes Camp focused on sharing an update on HWA biocontrol successes, and included a hemlock tree treatment demonstration from a certified pesticide applicator.

8.2 Watershed Data Scanning Project / Database Migration

Records of septic system designs dating back to the 1930's are archived in numerous file cabinets at the Skaneateles Gatehouse. As the Watershed Protection Program evolves and Watershed Rules and Regulations were updated, documentation collected on individual properties expanded to include all regulated activity involving county health departments, townships, and the NYSDEC. A database which records all activity in the watershed is maintained by city personnel and includes over 150 categories. The database is linked to a GIS database, which allows tracking of watershed activity such as new housing starts, violations of watershed rules and regulations, OWTS design approvals, etc.

In January 2010 the City initiated the scanning of all file folder documents. Electronic images will allow for indexing of data on individual tax parcels according to regulatory agencies, increase storage space, provide a back-up in the event of a permanent loss and allow for prompt retrieval through a GIS hyperlink. Scanning is conducted by City personnel and is typically scheduled for afternoons during the winter months when there are limited activity and access to property on the watershed. Due to the intermittent scanning schedule and extensive file folders, the project has taken numerous years. In the fall of 2025, the Watershed Protection Program initiated the process of upgrading its legacy watershed database. A development and consultation service provider was selected to assist with creating a new septic system management program. The migration to the new software program will enable the Watershed Protection Program to more efficiently identify, inventory, and manage onsite system compliance and service data.

8.3 Shotwell Brook Pathogen, Nutrient and Sediment Loading Reduction Initiative

Shotwell Brook is located along the northeast section of the Watershed and enters the Lake within proximity to one of the city's drinking water intakes. Agriculture accounts for approximately 70% of the land use in the 3.3 square mile Shotwell Brook Watershed. *The Land Protection Plan for the Skaneateles Lake Watershed* dated June 20, 1995, prioritizes Critical Management Zones, and divides the Skaneateles Lake Watershed into six Watershed Protection Zones. The Shotwell Brook tributaries and sub-watershed comprise the three highest levels of protection priority designated in the Report.

High intensity storm events frequently result in substantial sediment loading to Skaneateles Lake from Shotwell Brook. Storm events combined with strong southerly winds often result in sediment plumes which are transported from the Shotwell Brook outlet in the direction of Intake #1, causing elevated turbidity through the intake.

Shotwell Brook and Watershed Research, Field Assessments and Restoration Projects 2018 - 2024

2018: Stormwater Attenuation Projects

The SLWAP was awarded a $181,000 grant for storm water attenuation on the NE branch of Shotwell Brook in 2018. The project completed in September 2020 involved the establishment of a constructed wetland and an extensive floodplain on one acre of fallow agricultural land. An unexpected budget surplus allowed for a portion of the grant to be redirected to a floodplain restoration project located on the main branch of Shotwell Brook, east of the NYS Rt. 41 bridge. The restoration project involved the creation of four shallow pocket wetlands along 600 feet of Shotwell Brook. A rock cross vein was installed at the upstream section of the project to convey discharges above bank-full flow into the enhanced wetland, allowing for temporary storage and attenuation of peak flows. The project was completed in October 2021.

2023 – 2024: Shotwell Brook: Resilient New York Flood, Sediment and Debris Management Plan

In 2022 the New York State Department of Environmental Conservation (NYSDEC), in partnership with the New York State Office of General Services (NYSOGS), expanded the Resilient NY program by adding additional watersheds, including Shotwell Brook, with the purpose of creating more resilient communities. The objective is to provide communities with comprehensive basin wide flood and/or sediment and debris studies with actionable projects to improve the community's resilience. The Shotwell Brook watershed was selected as a priority watershed due to its proximity to Intake #1, and the watercourse being a major source of sediment to Skaneateles Lake. The objective of the study was to provide an effective method to identify areas within the Shotwell Brook basin where in-channel and overland sediment and debris transport contributed to excessive sediment release into Skaneateles Lake. A primary goal was to recommend projects that will reduce excessive deposition and debris, uncontrolled sediment sources, head cutting or downcutting of the channel, resulting in poor water quality in Skaneateles Lake.

An in-person project engagement meeting was held in November 2023 with involved agencies and local stakeholders to assist in the identification of high-risk areas. Field assessments and investigations of Shotwell Brook were conducted October 2023, and February 2024. The Final Report was submitted to the NYSDEC on September 19, 2024, identifying actionable projects within Shotwell Brook tributaries and watershed.

2024: Constructed Wetland Design & Permitting

In support of the implementation of a Resilient NY program recommended project on Shotwell Brook, the City of Syracuse and Skaneateles Lake Watershed Agricultural Program (SLWAP) entered into an agreement with a property owner in 2024 to create constructed wetlands adjacent to

Shotwell Brook. The 48-acre parcel includes approximately a 1,750 feet section of a Shotwell Brook tributary. The stormwater attenuation projects are located on abandoned agricultural land and an upland area that floods the owner's property during high flow storm events. Through NYS Legislative Initiative Funding (LIF) the City of Syracuse was awarded $100,000 for HABs abatement and a HAB Alert system in 2021. $75,000 of the grant was allocated for the design and implementation of the wetland projects.

2025: Completion of Constructed Wetland and Stream Restoration Project

Phase I of the project was the construction of a 1.2-acre wetland including an extensive floodplain allowing for greater storage during high flow storm events. This phase also included a new channel and stream meandering section upstream of the floodplain. The channel directs high flow events that exceed bank full into the new wetland, decreasing water velocity and allowing settling out of sediment. Wetland plants also remove and filter nutrients and sediments before stormwater flows are discharged over the downstream berm. Phase II was located upstream of the constructed wetland and involved creating 330 feet of meandering stream, restoring a severely incised section of Shotwell Brook. Restoring floodplain connectivity in this section is intended to reduce downstream water quantity and streamflow velocity during high intensity storm events.

2024: Cover Crops

In 2024 the SLWAP contracted with a private drone operator for the use of large aerial drone technology for planting seed on Skaneateles Lake Watershed farmland. Twenty-six acres of farmland along the Main Branch of Shotwell Brook was seeded with an annual ryegrass cover crop. Ryegrass was selected as a cover crop for its shade tolerance, allowing for germination and establishment under the canopy of corn leaves. Aerial seeding ensures a timely application that is often delayed with conventional farm equipment when extended wet weather patterns saturate fields, delaying corn harvesting and cover crop planting. An additional 48 acres of no-till drilled winter rye was applied to farmland along the Northeast branch of Shotwell Brook.

Focusing on a comprehensive multi-agency approach to reducing the amount of sediment discharging to Skaneateles Lake through Shotwell Brook has been and will continue to be a watershed protection priority.

8.4 Subsurface Agricultural Drain Tile Outlet Survey

In 2016 the Watershed Protection Program initiated a program to identify and record GPS coordinates of agricultural subsurface drain tile outlets throughout the Watershed. Watershed Inspectors are surveying every watercourse, road ditch, swale, grassed waterway, etc., that border agricultural fields. Outlet locations and observations noted within drainage ways such as erosion of embankments, excessive algae growth and foam which may be a result of subsurface discharges will be conveyed to the SLWAP for further analysis and possible modifications of Whole Farm Plans. Due to the limited time that can be allocated to the inventory, and the significant amount of land area to be covered (approximately 28 square miles of land use coverage in the watershed is agricultural) surveying all conveyances bordering farmland has extended over several years. In 2022, Watershed Inspectors completed the survey, logging a total of 227 drain tile outlet locations.

9. WATERSHED PERSONNEL TRAINING, CERTIFICATIONS AND PRESENTATIONS

The following workshops, conferences and webinars were attended by the Watershed Quality Coordinator in 2025. Presentations included OWTS's, wetlands, stormwater management and stream restoration.

- ***Septic Systems and Skaneateles Lake – August 21, 2025***
  Hosted by Cornell Cooperative Extension – Onondaga County
- ***NYSFOLA CNY Regional Conference – September 12, 2025***
  Hosted by New York State Federation of Lake Associations
- ***SLWAP Annual Meeting – December 11, 2025***
  Hosted by the Skaneateles Lake Watershed Agricultural Program
- ***Vegetation Management Strategies to Manage Erosion and Reduce Sediment – Baldwin County Conservation District – December 16, 2025***
- ***Streambank Erosion and Phosphorus Loading to Surface Waters – Sustainable Phosphorus Alliance – December 8, 2025***
- ***Riparian Restoration 101: Stream Restoration - Alliance for the Chesapeake Bay – December 15, 2025***
- ***Pulsed Stormwater Treatment and High-Flow Treatment of Stormwater – SERDP & ESTCP Webinar Series - Dec. 10, 2025***
- ***Messy Rivers are Healthy Rivers – Colorado State University December 17, 2025***
- ***Low-Tech Process Based Restoration and Bioengineering Techniques – WADE Virtual Conference - December 12, 2025***

The coursework and training listed above fulfilled renewal requirements for the Certified Professional in Erosion and Sediment Control (CPESC) through ENVIROCERT International Inc.

10. ACKNOWLEDGMENTS

The City of Syracuse continued in its efforts to control pollution within the Skaneateles Lake Watershed in 2025. It is with appreciation that the Water Department acknowledges the efforts, assistance and expertise of the Onondaga, Cayuga and Cortland County Health Departments and the NYSDEC Division of Law Enforcement and Division of Regulatory Affairs in the enforcement of the Watershed Rules and Regulations. In addition, the Water Department acknowledges the combined efforts of the City of Syracuse Department of Law, the New York State Department of Health, Natural Resources Conservation Service, Soil and Water Conservation Districts, and the Cornell Cooperative extension for their efforts in assisting the Water Department in its endeavors to enhance the watershed environment.

11. ENDNOTES/FOOTNOTES

1. Lake data excerpted from: Effler, S.W., et al. 1989. Limnological Analysis of Skaneateles Lake, 1988. Upstate Freshwater Institute, Inc. Syracuse, NY.

2. Historically, the City of Syracuse Department of Water has used 59.3 sq. miles or 37,952 acres as the watershed area when calculating the water budget. As of the current re-calculation of land use and cover, the GIS calculates the watershed area as 58.94 sq. miles or 37,720 acres with acreage for Onondaga, Cayuga and Cortland Counties at 19,310, 12,583 and 5,827 acres, respectively.

3. The data is based on the 2020 Census.

4. Average historical data is based upon City data for the 75 years between 1951 and 2025 inclusive, i.e., total inputs versus total withdrawals plus lake elevation changes.

5. There are two City operated rain gauges within the watershed. One is at the Water Plant in the Village of Skaneateles (Cooperating Observer for the National Weather Service) and the second is located at the southern end of the watershed in the Town of Sempronius on City owned property. Precipitation amounts referred to within this report are weighted values, i.e., 75% of the recorded amount at the Plant plus 25% of the amount recorded at Sempronius.

6. Volume is determined based upon a drainage area of 72.54 square miles.

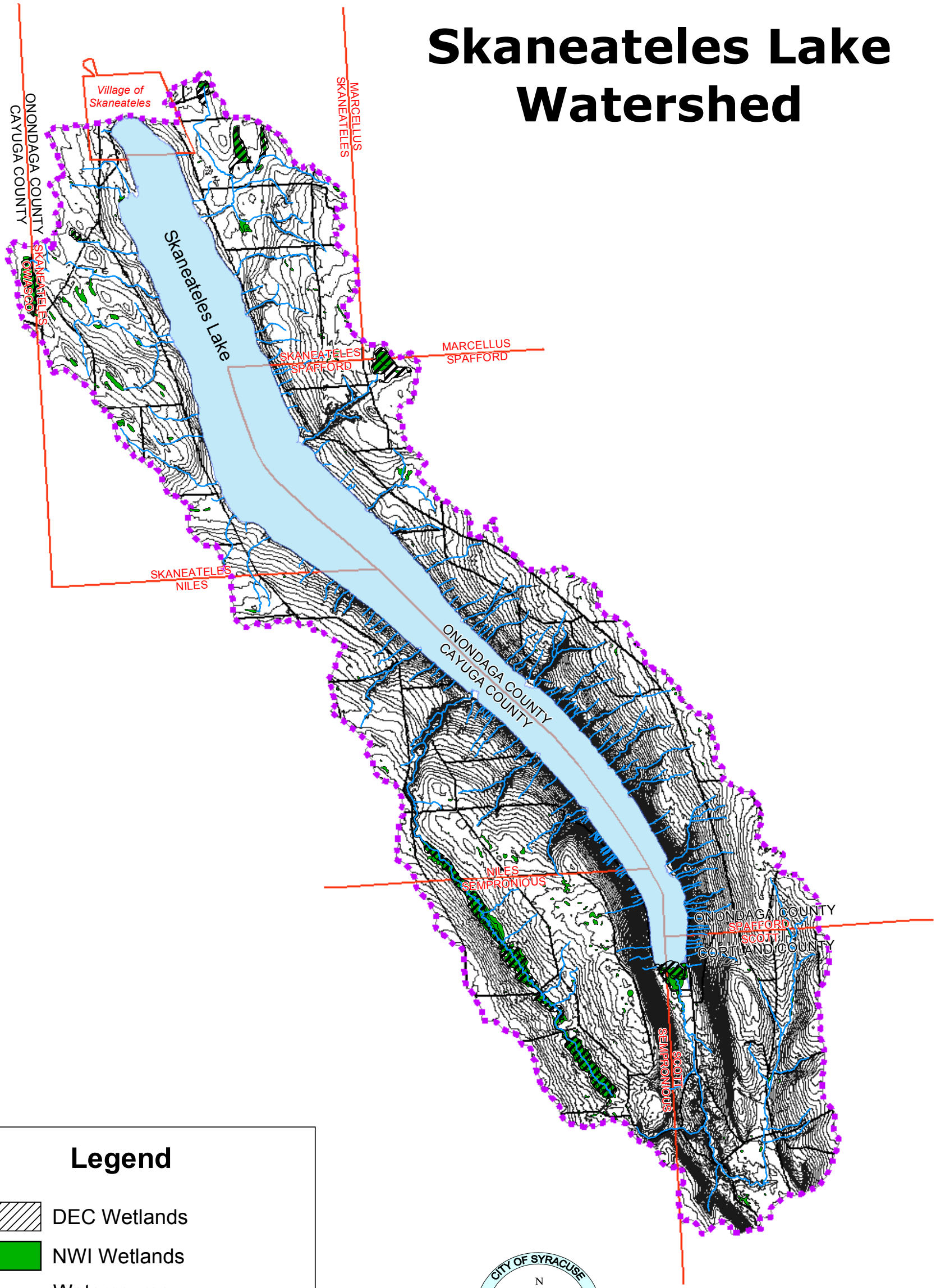
7. Based upon a lake surface area of 13.6 square miles.

8. This is required by Decision 609B, 1958 between the New York State Conservation Department/Water Power and Control Commission and the City of Syracuse

Appendix

Appendix A – Exhibits

Skaneateles Lake Watershed



Legend

- DEC Wetlands
- NWI Wetlands
- Watercourse
- Watershed Roads
- Municipal Boundaries
- Watershed Boundary
- Twenty-Five Foot Contour

CITY OF SYRACUSE
DEPARTMENT OF WATER
N
W
E
S

0 0.5 1 2 3 4

Miles

Exhibit A: General Watershed Map

Skaneateles Lake Watershed

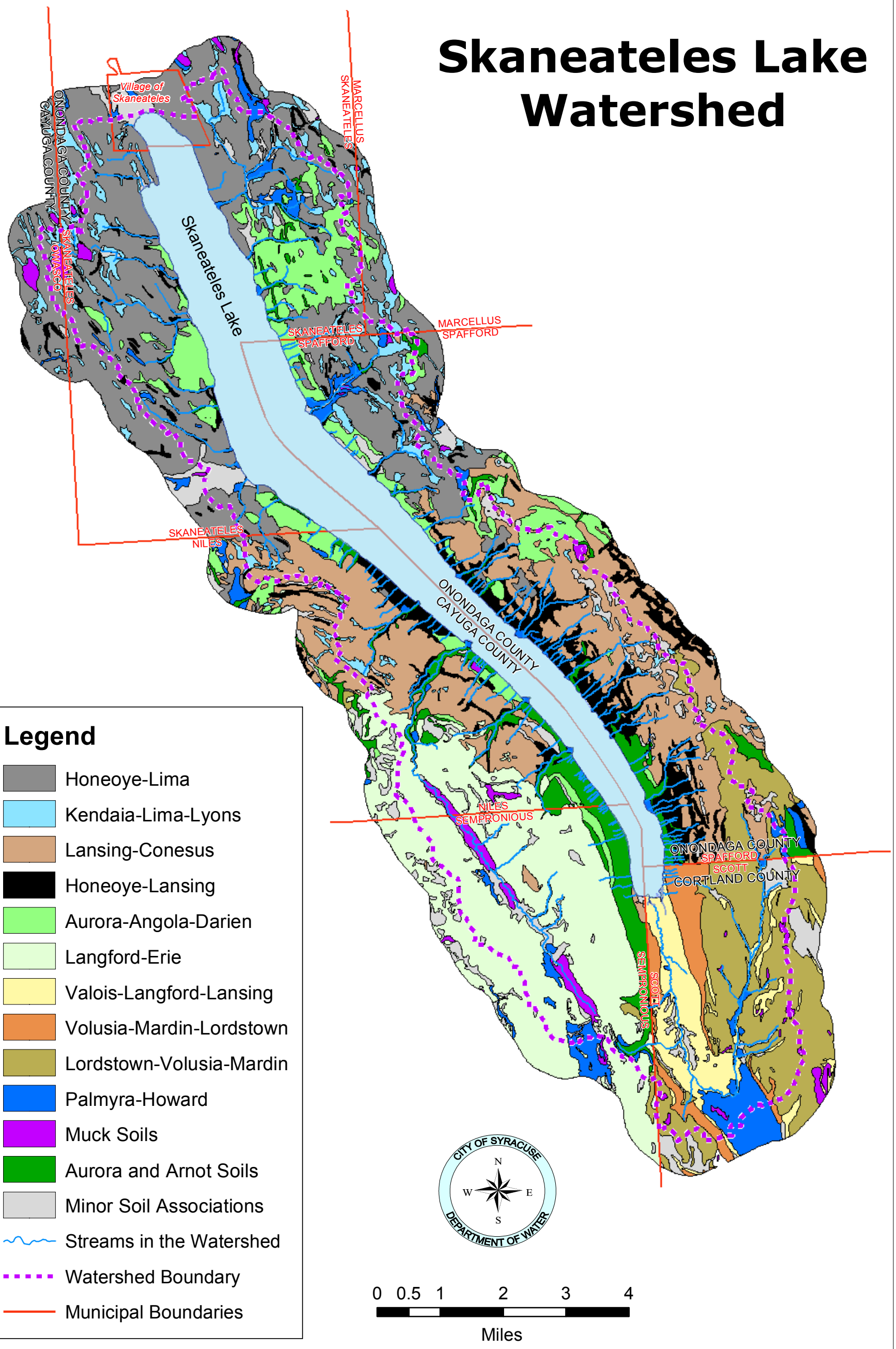


Exhibit B: Soil Associations Map

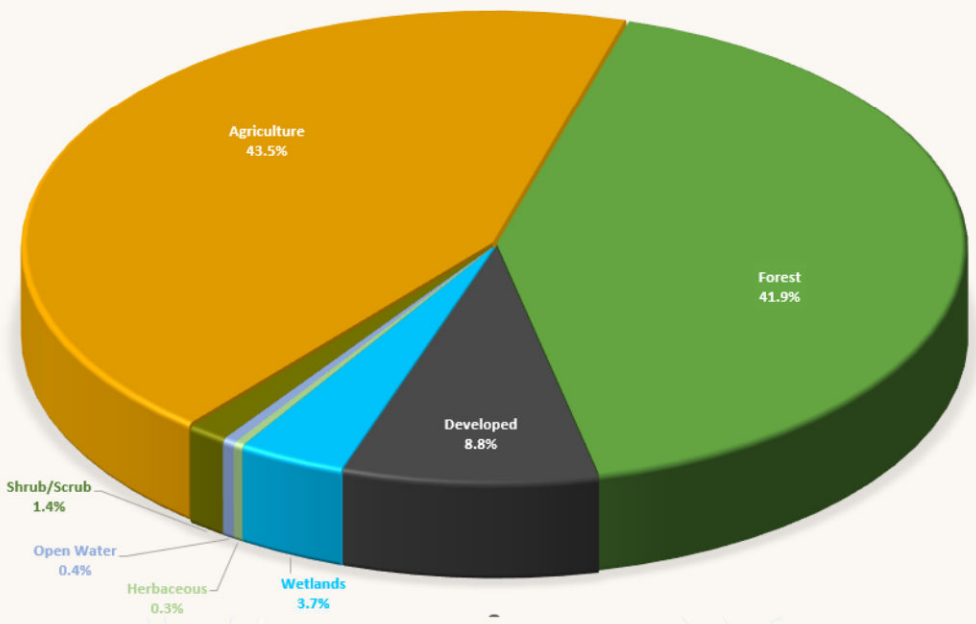
Skaneateles Lake Watershed

Skaneateles Lake

NICHOLS CORNERS
CLINTONVILLE
HEATH GROVE
Otisco Lake
WILLIAMS GROVE
OWASCO
NILES
OTISCO VALLEY
GLOBE HOTEL
SOUTH SPAFFORD
38A
COMO
HALLS CORNER

0 0.5 1 2 3 4
Miles

WATERSHED LAND USE PERCENTAGES
(Not Including Skaneateles Lake)



Data From 2021 National Land Cover Database

LAND USE BY COUNTY

	Cayuga County		Cortland County		Onondaga County	
	Acres	Percent	Acres	Percent	Acres	Percent
Agriculture	3978.3	31.8	1,831.4	31.3	10,588.3	54.7
Developed	871.9	7.0	376.0	6.4	2,058.2	10.6
Forest	6754.2	54.0	3,414.9	58.3	5,648.3	29.2
Herbaceous	18.3	0.1	11.1	0.2	93.9	0.5
Open Water	14.0	0.1	3.6	0.1	148.6	0.8
Shrub/Scrub	115.3	0.9	86.7	1.5	318.8	1.6
Wetlands	758.7	6.1	130.3	2.2	500.3	2.6
County Totals:	**12510.6**	**100**	**5,854.0**	**100**	**19,356.3**	**100**

Watershed Boundary
Watercourse
Municipal Boundaries
Roads
Land Cover 2021
Agriculture
Developed
Forest
Herbaceous
Open Water
Shrub/Scrub
Wetlands

Esri, TomTom, Garmin, SafeGraph, GeoTechnologies, Inc, METI/NASA, USGS, EPA, NPS, USDA, USFWS

Exhibit C: Land Use Map

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Syracuse
New York